Briefings to plant operators											
All Temporary Electrical Site installations and distribution systems shall as a minimum meet IP44 standards and be in accordance with:- (a) Indian Electrical Regulations; (b) The Power Companies' Supply Rules; (c) BS 7671 Requirements for electrical installation, the IEE Wiring Regulations (16th Edition); (d) BS 7375 Distribution of Electricity on Construction and Building Sites; (e) BS 4363 Distribution Assemblies for Electricity Supplies for Construction and											

Building Sites; and (f) BS 6164 Safety in Tunnelling in the Construction Industry.											
Monitor compliance daily											
All portable electrical appliance permanently numbers including sub-contractor											
Register of tools on site. Secure storage. Inspection of competent person & Authorised issue and return											
Rechargeable tools used preferentially.											

Task assessment sheets confirms condition check before use.											
Authorised users identified for electrical tools											
Manufacturer data used to assist in training											
Equipment in good condition. Evidence of equipment being in good order, clearly identifiable and being used correctly by trained, authorised person											
Good clear use of registers. Schedule of inspection periods displayed											
Maintenance schedule/hire status record											

available to monitor condition of electrical equipment on site and completed.											
Subcontractors provide records of inspection and procedure to remove faulty equipment found on site.											
Competent person C licence appointed for inspections and tests etc. Including Sub contractor activity											
Competent person identified on site notice board.											
Training records on site of competent persons.											
Permits contents communicated to operative,											

sign off procedure in place.											
Records maintained, regular monitoring and recorded inspection of permits											
Periodical cleaning and replacement of light fittings /lamps Pole height maintained as per legal requirements											
Light fittings installed & lay out plan ,pole number displayed and sufficient illumination maintained at site											
Register available for illumination and regular update											
Distribution & Electrics up to date and good records of											

corrective actions and changes recorded											
Records of weekly inspections and monthly audit.											
All work shall be supervised or executed by qualified and suitably categorised electricians.											
All cabling shall be run at high level whenever possible and firmly secured to ensure it does not present a hazard or obstruction to people and equipment.											
The installation on Site shall allow convenient access to authorised and competent operatives to work on the											

apparatus contained within.											
The following voltages shall be adhered to for typical applications throughout the distribution systems: (a) fixed plant - 415V 3 phase; (b) movable plant fed by trailing cable - 415V 3 phase; (c) installations in Site buildings - 240V 1 phase; (d) fixed flood lighting - 240V 1 phase; (e) portable and hand held tools - 110V 1 phase; (f) Site lighting (other than flood lighting) - 110V 1 phase; and (g) portable hand-lamps (general use) - 110V 1 phase.											

Earthing and bonding shall be provided for all electrical installations and equipment to prevent the possibility of dangerous voltage rises and to ensure that faults are rapidly cleared by installed circuit protection.											
Only plugs and fittings of the weatherproof type shall be used and they should be colour coded in accordance with the Internationally recognised standards for example as detailed as follows: (a) 110 volts : Yellow (b) 240 volts : Blue (c) 415 volts : Red											

Tunnel - The Contractor shall provide adequate lighting at the face and at any other point where work is in progress.											
Tunnel - Emergency lighting shall be installed at the working faces and at 100m intervals along the tunnel to help escape workmen in case of accidents.											
All lighting shall be kept maintained and clean at all times on the traffic diversion.											
Use a low voltage open circuit relay device if welding with alternating current in constricted or damp places.											

WORK IN CONFINED SPACES - Flame-proof lighting. (Hand lamps not more than 24 volts.);											
Procedure of LOTO											
Procedure of each type of equipment											
Standard maintenance procedure											
The Site mains voltage shall be as the Electricity Utility supplies, 415V 3-phase 4-wire system. (a) Single-phase voltage shall be as the Electricity Utility supplies, 240V supply. (b) Reduced voltages shall conform to BS 7375.											

Cables shall be selected after full consideration of the conditions to which they will be exposed and the duties for which they are required. For supply cables up to 3.3kV the cable armouring shall be used as the earth return in conditions where the cable is continuously extended and not subject to continuous movement after installation.											

For supplies to mobile or transportable equipment where operation of the equipment subjects the cable to flexing, the cable shall conform to one of the following specifications appropriate to the duties imposed on it: (a) BS 6708 flexible cables for use at mines and quarries; (b) BS 6007 rubber insulated cables for electric power and lighting; and (c) BS 6500 insulated flexible cords and cables.											
Provision of Silenced Equipment like D.G											

Specific Training Competency Certificate of electrical personals											
Admin & Stores											
As per the child labour abolistion act, are any body employed under the age of 18 years?											
Company / Organization registration Certificate available at the Admin?											
Contractor registration certificate available at office ?											
Employer Public Liability Certificate available and displayed											
Permission/Clearence to operate the Casting Yard											

Labour license available?											
Whether all the workmen covered under ESIC?											
Whether all the workmen covered under EPF Policy?											
Whether the workmen covered under group insurance policy?											
whether the project registered under BOCWR											
Whether the Contractor having the 'CAR' policy?											
Whether the minimum wages act implemented and being followed in practice?											
Pre-employement medical fitness is being carried											

out for all the workmen											
Workmen screening is being done as per their skills and trade											
Are Women deployed as per the guidelines of factories act / BOCW rules-2006											
Sexual harassment prevention is in place and policy formulated/dis played?											
Women protection act guidelines being implemented											
Workmen compensetion policy is available											
Emergency Contact Numbers displayed at suitable loctions.											

Ambulance is available											
Qualified Doctor is available											
Sufficient numbers of male nurses are available at all locations											
Health Centre is available as per the guidelines of BOCW act											
Bed and blankets in the Health Centre are available and are in hygienic condition											
Sufficient numbers of toilets are available at the project											
Sufficient quantity of drinking water facility is available											
Sufficient quantity of											

Urinals are available											
Separate washrooms with pictorial photos for wormen are made available											
Rest rooms for women are available											
Rain protection system is in place											
Mosquito prevention system during the rainy season is in place											
Fire Prevention/Protection systems are adequte at work place											
Labour camp set up is as per the legal requirement											
Separate kitchens in a safe distance for the workmen at											

Labour camp is provided											
Cleaning/white washing frency of Labour camps, toilets, urinals, wash rooms											
Fire Prevention/Protection systems are adequte at labour camp											
Periodic Mock drills at workmen camp											
Hand wash facility with soap water solution											
Drinking water test report											
Security arrangement at workmen camp											
Medical care / Doctor visit to Workmen camp											
Fumigation, Fogging,pest control, Anti larva treatment											

at Workmen camp											
Vaccination											
Transport facility for workmen from workmen camp to site.											
Drinking water tank cleaning frequency and other display posters											
Cloth washing facility at workmen camp											
Welfare, Health & Hygiene posters at workmen camp											
Health and Hygiene awarness programes at workmen camp											
Energy drinks, ORS,Lemon water, butter milk during the summer											

Emergency lighting arrangement is in place											
Stores											
License to stock Diesel & Gas Cylinders if applicable as per LR											
HIRA for stores											
Store Layout											
Stacking arrangemnts of material											
Manufacturer Test Certificates for lifting tools and tackles											
Calibration certificates for the measuring equipments											
Racks arrangement with safe distance											
Safe access / egress in side the store											

Fire fighting arrangement in the store											
Ladder provision to pick the material stacked at certain height											
Ventilation in side the store											
Illumination in side the store											
5 'S' system implementation											
Separate storage of flammable material and precautions											
First In / First Out system implementation for material											
Tag system for all material											
Emergency evacuation posters											
Emergency contact number display											

flame proof electrical fittings in the store											
First aid box											
Separate safe storage facility for wire ropes and web slings with test certificate											
Dispay of MSDS for all COSHH items											
Emergency lighting arrangemnt in side the store											
Drinking water facility											
Stacking arrangement of Hazardous material like bentonite etc.											
LED lighting arrangement to reduce power consumption											
Eye wash / Shower facility in case contact with hazardous chemical											

BUREAU VERITAS (INDIA) PRIVATE LIMITED MUMBAI

Bund wall / dyke wall for diesel, petrol, chemical starage area											
Mock drill for store											
Display of OH&S Policy in the store											
Display of Operational Controls in the store											
Entry restriction posters											
Informatory posters											
Dos & Don'ts display											
Smoke / Heat detectors											
Oil/Chemical storage in metal containers											
Waste disposal provision / plan											
Trainings for store staff											

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Calibration certificate for Weighing machine											
Loading / Unloading Procedures											
Wedge blocks / wheel chockers for parked vehicle at the store during loading / Unloading											

SECTION 2

GUIDANCE DOCUMENTS

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Construction Method Statements contents	GSAF 02		
Working At Height contents	GSAF 03		
OHS&E Training Matrix	GSAF 04		
Emergency procedure Fire/ Bomb Alert	GSAF 05		
OH&S Audit measurement Criteria	GSAF 06		
Emergency Response Plan contents	GSAF 07		
Health Plan contents	GSAF 08		
On Site Traffic Management Plan contents	GSAF 09		

Project OH&S Plan contents	GSAF 10		
Site Fire Plan contents	GSAF 11		
Traffic Management Plan contents	GSAF 12		
Training Implementation Plan contents	GSAF 13		
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LIFTING OPERATIONS CONTENTS

Rev.	Date	Description
0		Reviewed/edited/amended in conjunction with management review
0		Issued for use

	Prepared by:	Reviewed by:	Approved by:
Name:			
Date:			
Signature:			

Contents

GSAF 02

Construction Method Statement Contents

The headings and sub-headings listed below are not exhaustive	Identified Yes / No	

1.	**Purpose**		
2.	**Scope of Work (detailed)** Detailed description of Works to be undertaken. Limits of work and site boundaries, including time limits / Completion criteria		
3.	**References/Consents/Supporting Information**		
4.	**Hazards & Risks Identified** • List significant hazards with operation and as identified in the Project Safety Plan. • Consider health and safety implications • Risk Assessments • Hazardous substance assessments		
5.	**3rd Party/Arrangements/Protection/Communication & Liaison** **Method** • Movement – storage of materials and equipment • Restricted clearances – plant & equipment • Reduced site lines for travelling public and construction plant • Occupied premises within or adjacent to operations • Over-flying work operations (cranes) • Site security • Road traffic management		

	• Pedestrian management • Interface with public bodies and schools • Public safety • Temporary fencing and protection **Communication & Liaison** • Identify specific persons who must be contacted, additional to those specific in the Project Safety Plan e.g. other section engineers • Consider other contractors working nearby. Highway Authorities. Occupiers of adjacent property. Client undertakings. • Sub-standard conditions reporting **Supporting Information** Drawing and layout of initial, interim and final works. Temporary Works design, support calculations, checking and approval Quality control arrangements		
6.	**Site Spatial Management Plan** Site Layout Plan shall include the following • Worker's Passage Route • Vehicle Running Route • Construction work area • Workshop area • Material storage area • Construction waste storage area		

7.	**Environmental** • Noise, dust, smoke, mud, vibration. • Disposal of waste, frequency and method of disposal. • De-watering arrangements and disposal of water • Pollution controls Fuels, oils etc., storage and containment Environmental – liaison		

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8.	**Method: Briefing Arrangements, Plant, Personnel** **<u>Method</u>** • Reference to programme chart showing sequence of separate tasks • Standards and Procedures • Sketches • Access and egress arrangements • Delivery of materials • Details of temporary structures • Method of authorising start of work • Risk Assessments considering Health and Safety **<u>Personnel</u>** • Number of contractors and sub-contractors • Communication methods • Supervision arrangements, including names of person in charge • Competence and training requirements. (especially in respect of plant and equipment used) • Working hours • Shift hand-over arrangements • Welfare and first aid • Access requirements, special conditions **<u>Briefing Arrangements</u>** • Determine level and extent of briefing arrangements including accompanying documentation. • Detail how understanding is to be confirmed by contractors and or individuals **<u>Plant & Equipment</u>** • Specify plant and equipment to be used • Competence requirements to operate or erect plant and equipment • Authority to work • Operational Restrictions		

	• Permit systems • Inspection and examinations • Record keeping • Temporary lighting • Detail of cranes, lifting machines, appliances and lifting tackle; including details of site access, • Rigging and de-rigging		
	Infrastructure Protection • Identify hidden services • Use of approved Cable Locating Tools • Identify infrastructure susceptible to damage e.g. power and telecommunications equipment, • pipes, air mains, fire detection equipment • Sketch showing location of services or reference where information can be found. • Isolation and protection – safe working locations. • Permits to work • Lock out procedure and control. • Hand over and re-energising • Plant movements. • Additional fire precautions • Hot Work arrangements • Structural considerations for existing buildings & structures.		
9.	**Quality** • Quality Control arrangement		
10.	**Safety** • Control measures for specific health hazards e.g. Leptospirosis • Relevant contractor's risk assessments, including COSHH and manual handling • Permit to work systems • Personal protective equipment requirements • Etc		
11.	**Hold Points** Hold Points are trigger levels if exceeded then the work ceases. This will include safety, quality and environmental issues.		
12.	**Inspection & Test Plans**		

WORKING AT HEIGHT CONTENTS

Rev	Date	Description

		Reviewed/edited/amended in conjunction with management review
		Issued for use

	Prepared by:	Reviewed by:	Approved by:
Name:			
Date:			
Signature:			

OHS&E Training Matrix

	Management																				Supervisor											Specific												
Types of training	OHS&E Orientation	OHS&E Leadership	OHS&E Plan	OHS&E Improvement Plan	Management of Change	OHS&E Audit & Inspection	OHS&E Emergency Response & Preparedness	Incident/Accident Investigation & Reporting	OHS&E Communication	OHS&E Promotion & Incentives	Traffic Management	Hazard Identification & Risk Analysis	Permit to work system	Confined space entry	scaffolding	Waste Management	Environment Monitoring	Labour welfare measures	Behavioural Based Safety Management (BBSM)	Job/Task Safety Analysis (JSA)	Safety Training Observation Programme (STOP)	Industrial First Aid & CPR	Incident / Accident Investigation & Reporting	Fire fighting	Confined Space Testing & Certification	Scaffold Erection & Inspection	Rigging	Wire Rope Inspection	Crane Inspection	Electrical/Mechanical Isolation	Permit to Work System	Confined Space Working	Explosive Handling & Control	Heavy Lifting Operation	Radiography (X-Ray)	HAZMAT Handling & Control	Welding, Cutting & Bracing	Power Actuated Hand Tool	Electrical/Mechanical Isolation	Roofing Work	Steel erection work	Scaffold Erection/Dismantling	False-work Erection / Dismantling	Painting in Confined Area
Project Director /Project Manager	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•		•																						
Construction Managers/Engineer	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•		•					•								•									
Quality Manager/Engineer	•	•	•	•	•		•	•	•		•	•	•	•	•			•	•	•	•	•																						
Planning Manager /Engineer	•	•	•		•	•	•	•	•			•		•								•																						
Public Liaison Officer /Labour Welfare officer	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•				•	•	•	•	•													
Construction Supervisors /Foreman	•		•	•	•	•	•	•	•	•	•	•	•			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Senior/Junior Electrical Manager / Engineer	•		•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Machinery Operators	•						•				•					•						•		•			•																	
Material Handlers	•						•					•	•		•							•	•	•			•																	
Station Building Workers	•						•						•		•	•						•		•							•			•		•		•		•		•	•	•
Steel workers	•						•						•		•	•						•		•			•				•			•		•	•			•	•	•		
Mechanical workers	•						•									•						•		•			•				•	•		•		•			•	•		•		•
Other Civil workers	•						•									•						•		•			•				•	•	•	•		•			•	•		•	•	•
Electrical /Plant Foreman /workers	•						•									•						•		•			•				•	•		•		•			•	•		•		•
Radiographers	•						•									•						•		•							•	•			•	•						•		
Transportation Drivers	•						•				•					•						•		•																				
Traffic Marshal /Security Officers	•						•	•			•	•	•		•	•						•	•	•																				
Traffic Engineer /Barricade Manager /Housekeeping Manager	•		•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Medical Doctor/Male Nurse/First Aiders	•	•	•				•	•							•	•		•				•																						
Chief /Sr. OHS&E Managers		•					•								•		•					•																						

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	Management																				Supervisor											Specific												
Types of training	OHS&E Orientation	OHS&E Leadership	OHS&E Plan	OHS&E Improvement Plan	Management of Change	OHS&E Audit & Inspection	OHS&E Emergency Response & Preparedness	Incident/Accident Investigation & Reporting	OHS&E Communication	OHS&E Promotion & Incentives	Traffic Management	Hazard Identification & Risk Analysis	Permit to work system	Confined space entry	scaffolding	Waste Management	Environment Monitoring	Labour welfare measures	Behavioural Based Safety Management (BBSM)	Job/Task Safety Analysis (JSA)	Safety Training Observation Programme (STOP)	Industrial First Aid & CPR	Incident / Accident Investigation & Reporting	Fire fighting	Confined Space Testing & Certification	Scaffold Erection & Inspection	Rigging	Wire Rope Inspection	Crane Inspection	Electrical/Mechanical Isolation	Permit to Work System	Confined Space Working	Explosive Handling & Control	Heavy Lifting Operation	Radiography (X-Ray)	HAZMAT Handling & Control	Welding, Cutting & Bracing	Power Actuated Hand Tool	Electrical/Mechanical Isolation	Roofing Work	Steel erection work	Scaffold Erection/Dismantling	False-work Erection / Dismantling	Painting in Confined Area
Jr.OHS&E Managers/ OHS&E supervisors /Stewards	·		·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
Environmental Manager /Engineer	·		·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·

EMERGENCY PROCEDURE FIRE/BOMB ALERT

OPERATING COMPANY:			
SITE / CONTRACT:			
SITE ADDRESS:			
SITE TELEPHONE No:			

FIRE / BOMB WARDEN DETAILS

NAME	AREA COVERED	CONTACT TEL	POSITION

PROCEDURE

1. A record of all personnel must be kept on site each day.(Visitors Register and the Operatives Daily Register).
2. All personnel must be made aware of the fire drill during the Safety Induction.
3. All personnel must familiarise themselves with the location of Fire Extinguishers Fire Alarm Call Points (temporary and permanent), Escape Routes and the Designated Assembly Point.
4. If a fire or bomb is discovered warn others by shouting and ensure TEL 999 is called.
5. Send a colleague to advise the designated fire wardens /fire safety co ordinator/site manager.
6. If it is a fire raise the fire alarm at a Break Glass or by sounding any temporary sirens / horns/ bells supplied for this purpose.
7. Try to put out the fire only if it is safe to do so, using the appropriate fire fighting equipment.
8. On hearing the fire alarms evacuate the workplace leave tools, equipment and personal belongings behind.
9. Proceed to the Assembly Point where a check will be undertaken to confirm all personnel are out.
10. Inform the fire/bomb warden or those in authority if you know someone is missing.
11. Do not return to the work place until instructed by Rok Management.

NOT PUT

In order

If you do

52.18 Operating procedures

52.18.1TBMs with man-entry facilities to a normally flooded slurry chamber will require extensive procedures to ensure safety for personnel, These procedures shall be developed by the contractor and submitted to the Employer / Engineer for notice of no objection. It is usual to employ compressed air to maintain face support. The following safety precautions must be employed.

a) No person should enter alone.
b) Communications are essential and a plug-in type intercom is suggested.
c) Air quality should be monitored.
d) There should be a recognized method for removing injured persons from the front of the machine and airlocks should be able to accommodate a stretcher.
e) During work on the TBM, a constant watch should be kept to detect deterioration of the face.
f) If conditions appear unfavourable, the crew should be withdrawn and the chamber reflooded. Simultaneous maintenance should not be carried out on the rest of the system.

52.18.2The slurry plant, pumping system and TBM must be kept in a state of readiness in order to be able to excavate one cycle so that tunnelling forward into good ground can quickly recommence.

52.18.3High pressure on the face of a TBM during driving can result in high pressures at the rear of the shield. This can affect the installation of segments and can present hazards to segment-handling personnel. There shall be a procedure to identify when such difficulties occur so that suitable reduction of the slurry pressure on the face of the machine can be carried out.

52.18.4During the launch of a shield from a shaft, it is possible that high slurry pressure will cause discharge into the shaft. It is essential to manage any leakages and contain any pressures so as not to cause a loss of ground and slurry into the shaft or the separation system, with the attendant risks to persons and the security of the works.

52.19 Slurry separation

52.19.1Slurry separation can be carried out by the use of settlement tanks, separation plant or lagoons. Tanks must be secured safely with adequate foundations and support. Where there is any risk of falls into a tank, the tank must be protected with grating. Where separation plant is employed, in addition to the protection of tank facilities, all moving parts must be guarded. An isolator shall be installed to enable a qualified operator to shut down the plant.

52.19.2As with any fluid handling, matters can go wrong quickly, and emergency procedures must be established in advance. All maintenance and adjustments shall be carried out in accordance with manufacturer’s instructions. Where lagoons are employed, there must be adequate fencing, determined by risk assessment, and warning signs identifying the depth of the lagoon and the nature of the contents.

52.19.3Removal of slurry from a site must be managed with due recognition of the nature of the contents. Slurry must be pumped into suitable tanks and tankered off site to a licensee with waste transfer notes maintained by the contractor. The contractor and their suppliers are subject to auditing by the Employer / Engineer

52.20 Tipping and disposal

52.20.1The disposal of excavated materials off site is subject to local and national waste disposal legislation.

52.20.2The location of any on-site tip or disposal area must be carefully chosen and due consideration given to engineering requirements for tip siting and formation, e.g. method of construction, drainage and compaction. It should not be positioned to surcharge future tunnel excavations unless it can be established that the soil or rock is able to withstand these additional loads. In tipping areas, reasonably level and properly maintained roadways should be provided using ballast or hard-core, as necessary, to allow proper traction and stability of haulage and tipping vehicles. If tipping is to be carried out down a slope, a robust “stop” should be provided to prevent any overrun of the haulage vehicle.

52.20.3Dust control measures shall be put in place during the tipping and storing of rock and spoil to minimize risks to health and the environment.

52.20.4It is often necessary to provide facilities for cleaning wheels, etc., before the vehicles go on to the public highway. The legal requirements for this are generally the subject of local by-laws. A further hazard with pumped slurry storage is the possibility of flooding. The flood protection at each shaft must be well above any possible slurry lagoon level.

52.21 Transport and loading

52.21.1Track maintenance should not be attempted during train operations. A safe system of work should be adopted which includes adequate refuge space, a nominated lookout, and flashing warning lights being placed at either end of the section of track affected.

52.21.2Locomotive drivers should be made aware of the work in progress.

52.21.3Locomotive drivers should be authorized in writing by the site management after their training, testing and the issue of a “Authorized certificate”.

53. Tunnel plant

53.1 General

53.1.1Due to the confined space in a tunnel, there is a greater likelihood of hazards occurring. Safety aspects specific to tunnelling. Before attempting any work on powered machinery, it must be isolated. This will apply equally to electrical and mechanical plant, and suitable instruction and training must be given, backed up by notices on the machinery advising on the steps to be taken. The use of burning and welding equipment in a tunnel shall be minimized and cylinders, etc. should be transported in equipment specially designed for that purpose.

53.2 Pneumatic plant

53.2.1Flexible hoses are a hazard if they fail in use due to the violent whiplash effect caused by escaping air. They must be protected and routed clear of possible impact or cutting damage. It is good practice to rate the hose at a higher pressure than anticipated working pressure to allow a reasonable margin for abuse and attrition. Typically, 10 bar hose is used in 7 bar applications. The hose end fasteners must either be banded by a trained person or secured with correctly matched bolted clamps. It is important to be aware that several different outside diameters are available for a given internal diameter of hose, and different clamps will apply. Where hoses are 25 mm internal diameter or larger, whip restraint wire loops must be attached at each end. All hoses shall be regularly inspected at least weekly for damage or deterioration and must be discarded if there is evidence of damage from oil (a swelling and softening of the rubber material), blistering, chafing, or cuts.

53.2.2Air-driven machinery, whether percussive hand tools, air motors, or reciprocating pumps, can be unacceptably noisy if used un-silenced. The level of noise produced by un-silenced motors prevents effective communication and is a hazard. Silencers must be fitted, of a type not prone to “freezing up” due to excessive restriction. The exhaust air shall be directed away from the operator. If the air supply is very wet, water traps should be used in the air line feeding the plant to minimize exhaust mist.

53.3 Hydraulically operated plant

53.3.1Fire on hydraulic power packs and hydrostatic systems can burn fiercely and be extremely difficult to extinguish with hand-held appliances. The following precautions shall all be taken as far as practicable.

a) Fire-resistant or low flammability fluids should be used in the system where possible. If mineral oils are essential, these should have a minimum flash point of 250 ºC.

b) Effective oil coolers should be incorporated to maintain temperatures below 70 ºC. Temperature switches should automatically shut down overheated circuits.

c) Fixed fire extinguishing systems should be installed covering major systems such as oil storage tanks, pumps, motors, etc.

d) Water curtains or other smoke suppression systems should be fitted. Circuit designers should consider the most appropriate type of pipework for an installation. Compression fittings on rigid pipes can suffer from loosening caused by circuit hammer and vibration, while flexible hoses are prone to chafing if not adequately secured.

e) Pipe runs should be routed to minimize the possibility of oil sprays reaching hot objects.

53.3.2Hydraulic systems should be regularly maintained, at a frequency determined by project-specific duties and hazards, to prevent leaks and removal of damaged pipework. Hoses should be pressure-rated to allow for transient pressure spikes in use.

53.3.3Motors used on segment erectors must be fitted with fail-safe brakes to hold loads in event of power failure or leakage. All hydraulic cylinders supporting loads that could cause injury if released shall be fitted with load holding valves. prEN 12336 gives requirements for vacuum pad systems on erectors. Face and forepole cylinders should be able to relieve hydraulic pressure automatically during shield shoving to avoid overpressure in the cylinders.

53.4 Internal combustion engines

53.5 Diesel engines

53.5.1Engines of more than 37 kW rated output shall conform to BS EN 1679-1:1998. This is to prevent the emission of exhaust gases that will cause excessive levels of particulates in the tunnel (i.e. more than 500 ppm9) CO, CO2, 750 ppm NO and NO2, 200 ppm hydrocarbons), and to ensure that the engines are constructed not to be a fire hazard. Ventilating air must be sufficient to dilute toxic gases to safe levels, and reduce smoke and odours to acceptable levels.

53.5.2To avoid local concentrations of dangerous levels of fumes, some form of exhaust conditioning must be used. This can take the form of a fume diluter or catalytic converter. Advice on correct matching shall be sought from the suppliers of any conditioning device. Diesel particulate filters can be incorporated, but the duty cycles on locomotives can cause fouling up. Engines for underground use must be "clean burn" types, producing minimum fume emissions, with a virtually clear output. Black or brown smoke indicates un-burnt fuel caused by over-fuelling and poor combustion, whilst white smoke indicates an engine in need of attention. Blue smoke caused by lubricating oil consumption indicates a worn engine. In addition, adequate provision must be made for ventilation and for the storage and handling of fuel. Explosion-protected equipment shall be used whenever there is danger of the presence of explosive gases e.g. methane.

53.6 Siting of engines

53.6.1Internal combustion engines at the surface must be sited so that exhaust fumes cannot enter the ventilation system or compressed-air intakes, or enter the tunnel by any shaft or other opening.

53.7Concreting plant

53.7.1 Pressure concreting pipes must be installed and maintained to pump manufacturers' guidelines. It must be remembered that these pipes can move with the pump strokes and that reaction forces are generated at every bend. Furthermore, the full pipeline will be heavy and must be supported, particularly in shafts. Wear on bends and reducers can cause complete failure under pressure and these areas should be checked regularly and records maintained.

53.7.2 When "blowing out", the cleaning ball or sponge can exit with considerable velocity and should be caught in a basket at the end. Once the ball leaves the pipe, a reaction "kick" from the escaping air can dislodge the pipeline, particularly at flexible hoses or bends. As an alternative to pneumatic blowing out of pumping lines, the system can be cleaned by washing out with water. The pipework used with the concreting plant should be correctly matched to the performance of the pump and its condition regularly checked. Incorrect pipe and couplings could rupture in use. Procedures shall be approved in advance for dealing with blocked pipelines as this will entail splitting and blowing out the line section by section.

53.8 Water drainage pumps

53.8.1Reliability of the drainage system, and adequacy of pumps, is critical for the safety of personnel where flooding could occur. A risk assessment shall be carried out and a system designed to eliminate the flood risk as far as reasonably possible. Consideration must be given to duplicate pumps and pipework, and to good accessibility of pumps for regular servicing or changing over. Submersible pumps should be suspended so that they can be progressively lifted clear of silt in the pump sump. Pump performance must be checked regularly to detect wear or blockage. Pipework in tunnels is prone to progressive silting up in use. A second pump line could be used to allow cleaning of the blocked pipes.

53.9 Drilling and piling rigs

53.9.1Drilling and piling rigs used in any operation associated with tunnel construction should conform CHAPTER XXIII of TBOCWR 2006 or BS EN 791:1996 and BS EN 996-1995 +A3:2009 respectively. All other drilling and piling "in-hole" equipment should conform to the appropriate International, European or national standard.

53.9.2The operation of drilling or piling rigs and their ancillary equipment should be carried out in accordance with established safe working practices and OH&S Conditions of Contract. Drilling and piling rigs must be maintained in accordance with the instructions in the manufacturer's handbook.

53.10 Specific recommendations for drill rigs

53.10.1The rig and in-hole equipment selected for a specific operation must be suitable for its intended use. Consideration must be given to site, operational and environmental conditions. Some drilling operations require access to the area around the rotating drill string. The operator must therefore be particularly careful in controlling the drill, when persons are working near the drill string.

53.10.2Machine ratings must be compatible with anticipated drilling conditions to prevent overloading of a machine. The drill rig shall be set up so that it forms a stable, safe drilling platform to reduce the risk of slipping or falling and so that all parts of the equipment are secure. Any limitations on ambient temperature for which the rig is designed should also be adhered to.

53.11 Grouting equipment

53.11.1Grout mixers employing rotating paddles must be designed so that any dangerous parts are either out of reach, or guarded. Before the removal of the guard, the power source must be isolated under a permit to work system e.g. lock-out. Grout pumps used for either transferring bulk grout or injecting at the point of use should be matched to the method of work. The potential pressure of some piston pumps is very high (66 bars) whereas air-operated diaphragm pumps will produce modest pressures (6 bars). It is essential that pressures be monitored and all hoses, pipe fittings and injection nozzles (grout guns) be matched to the maximum pressure rating of the pump and inspected regularly for damage and wear. High-pressure grout blow-outs can cause serious injuries. Where transfer distances are great, an effective means of communication shall be in place between the source and the point of delivery.

53.11.2The compressed-air displacement grout pan is a pressure vessel and is therefore a potentially dangerous piece of equipment, and must only be operated only by a person who has been instructed in its use. Opening of the lid under pressure must not be possible. All grouting equipment should be regularly inspected and maintained and any grout build-up removed. Whip checks should be provided for both the feed air and the grout transfer hoses. Potential hazards arising from grouting equipment include the following:

a) bursting of the grout feed hose through damage, inadequate maintenance or improper connection, or solidification of the grout. Cleaning and maintenance to a high standard is therefore essential;
b) blow-out at the point of injection where a screw connection or other pressure retaining device should have been provided;
c) back-flow of grout after injection. A stop valve may be necessary to retain the grout until it has set;
d) damage to lining or to surface installations due to excessive pressure.

54. Maintenance and repair

54.1 Inspection and assessment

54.1.1Detailed inspection and recording of the tunnel lining details should be carried out at planned regular intervals. A formal reporting procedure must be established with appropriate checklists to enable correlations with subsequent inspections. Before each inspection, reference should be made to the records of any previous inspection and behaviour, copies of which should be retained and available on site. Points which are likely to require attention are:

— Degradation and spalling of the lining material or exposed rock faces;

— Deformation and cracking;

— Location and quantity of ingress of water and solids;

— Chemical and thermal effects.

54.1.2In certain tunnels where the presence of gases poses a danger, a scheme must be drawn up for the regular assessment of the gases. Those areas of the tunnel lining that have deteriorated the most should undergo more detailed inspection and monitoring. In addition to normal inspection procedures, specialized techniques that can be employed include:

a) CCTV inspection;
b) Installation of monitoring studs for the accurate measurement of deformation;
c) Strain gauges;
d) Profiling, e.g. photogrammetry; laser measurement techniques;
e) Water sampling;
f) Geophysical, e.g. ground penetrating radar, thermography.

54.2 Preparation for repair

54.3 Site investigation

54.3.1It is important to carry out a site investigation to identify the ground conditions (if this is not already known) and the extent of the voids or defects behind or within the tunnel lining. The location and delineation of voids behind the tunnel lining is vital and will need to be determined. Trial holes could be required to establish, for instance, the general integrity of the tunnel lining in the area of the proposed work, whether a structural invert is present or to establish the groundwater regime behind the lining.

54.4 Design

54.4.1The design of the existing structure should be assessed for any changes in the ground or tunnel lining since it was built. The engineering properties of the ground or of the structure itself could have changed as a result of the action which has prompted repair (e.g. water leakage, solids transport, general weathering effects); and this could have affected the structural behaviour and load-bearing capacity of the ground/structure complex. The work to be carried out can itself impose local or widespread loading on the structure (e.g. grouting pressures). Even minor intrusions, such as drilling for grouting, can affect structural behaviour. Major intrusions, e.g. for partial reconstruction, will have greater effect. The design assumptions made and the limitations on working must be clearly stated in the health and safety plan, so that the contractor is fully aware of the dangers and able to design appropriate temporary works and emergency procedures.

54.5 Preparation of a safe system of work

54.5.1Detailed method statements, developed from the aspects and impacts associated with the task must be prepared for the tunnelling work. Statutory authorities and other bodies should be consulted as appropriate and their consent obtained if necessary. Method statements are to be submitted to the Employer / Engineer for review, notice of no objection and or no objection.

54.6 On-site procedures prior to repair documents to be held on site

54.6.1The principal contractor must ensure that site-based representatives of all parties to the contract have access to the following documents:

a) Health and safety plan and method statements with access to the health and safety file;
b) Relevant “as built” drawings;
c) Details of the latest condition survey;
d) The site investigation report;
e) “Permit to work” forms as appropriate;
f) Relevant parts of any by-laws, wayleave stipulations, etc.;
g) Other reports, drawings, etc. located during the historical research.

54.7Emergency procedures

54.7.1Before repair work commences, a site meeting of all relevant parties must be held to discuss emergency procedures. Further meetings shall be held as necessary during the course of the work.

54.8 Ventilation and testing for gases

54.8.1The procedures for ventilation during construction are equally applicable to repair works. Gases, particularly methane, can fill voids behind tunnel lining or enter the tunnel dissolved in groundwater.

54.9 Work in shafts

54.10 Lifting operations

54.10.1Cranes can be used for raising or lowering inspection cages but must not be used for supporting working platforms. Particular care shall be taken to prevent sudden crane movements.

54.11Environment

54.11.1Following the completion of work carried out within a shaft, but before re-entry, the environment within the shaft and within the tunnel at the bottom of the shaft must be checked to ensure that it is still safe and the check recorded. It should be noted that closing a shaft can result in a changed environment within the tunnel and therefore due regard must be given to ventilation, atmospheric pressure effects, means of escape, etc. This can also result in a shaft becoming a confined space. Build-up of water and debris on the temporary cover, decking or working platform must be controlled.

54.12 Access control

54.12.1Toe boards and hand rails shall be installed to control access to the top of the shaft and to prevent materials or debris falling on to persons working in the shaft. For work of limited duration within the shaft, access to the tunnel at the bottom of the shaft must be isolated. For work of longer duration or where necessitated by the nature of the work being undertaken, it may be necessary to install a temporary cover, decking or working platform within the shaft. This will generally protect the area within the tunnel at the bottom of the shaft. Rope access techniques can be used for access within shafts, where appropriate however this can only be carried out by specialists in rope access.

54.13 Working platforms

54.13.1Any working platform must have a substantial roof with a trap door. This will give protection against wet conditions and any stones falling down the shaft. The working platform shall be secured to the shaft wall during use and the maximum platform loading clearly displayed. Platform suspension systems must be designed to withstand the heaviest loading to which it is envisaged that the platform will be subjected. A secondary escape facility from the platform shall be provided. A means of communication with the working platform or inspection cage must be in place at all times. Communication should be both oral and visual wherever possible. Unless working from a properly guarded fixed working platform, suitable individual safety harnesses must be utilised.

54.14 Temporary works

54.14.1Before work on site commences, the contractor should submit to the Employer / Engineer for notice of no objection all details of temporary works, including proposed methods of working.

54.15 Record of work

54.15.1All work carried out should be recorded in the OH&S file so that it can be available for future inspection and repair work.

55. Radiation

55.1.1The use of radioactive substances and radiating apparatus shall comply with the Govt. regulatory requirements.

55.1.2Operations involving ionising radiation shall only be carried out after having been reviewed without objection by the Employer / Engineer and shall be carried out in accordance with a method statement.

55.1.3Each area containing irradiated apparatus shall have warning notices and barriers, as required by the Regulations, conspicuously posted at or near the area.

55.1.4Radioactive substances will be stored, used or disposed shall be strictly in accordance with the Govt. regulatory requirements.

55.1.5The contractor shall ensure that all site personnel and members of the public are not exposed to radiation.

56. Fire and smoke

56.1 Storage of materials

56.2 General

56.2.1Effective fire prevention depends on the elimination or control of materials likely to ignite or assist in the development of fire. In the confined space of a tunnel, strict precautions are essential. The amount of combustible material (e.g. timber, straw, paper, rubber), flammable liquid (e.g. oils, chemical solvents and primers, paraffin) and compressed gas in a tunnel should be kept to a minimum. Any materials not required imminently (generally those not needed during the working shift), shall be removed to a surface storage area, except items such as emergency face timbers and, in some cases, straw needed for safety purposes. Smoking and the carriage of smoking materials constitute a fire hazard, and shall be prohibited below ground and only permitted in designated areas on the surface away from combustible materials.

56.3 Combustible materials

56.3.1Combustible materials stored in a tunnel must have suitable fire-fighting equipment conspicuously located nearby (see Table 5 and Table 6). Storage areas for combustible materials shall have well-placed notices warning of the fire risk. Whenever practicable, the storage areas shall be isolated from the general working area. They must not be in, or in the vicinity of, any shaft or tunnel opening, or on any escape route, and be protected by some form of construction that has a fire-resistance period of not less than 30 min when tested in accordance with BS 476-20, BS 476-21 and BS 476-22. Surface structures and buildings at, or close to, any shaft or tunnel opening should be constructed from non-combustible materials wherever practicable.

Table 5 — Provision of fire extinguishing equipment

Location of fire	Extinguishing medium				
	Water (jet)	Water (spray)	Foam	Inert gas	Powder
Tunnel — general	**F**		**P**	**P**	**P**
TBM — general			**P**	**P**	**P**
TBM — hydraulics			**F**		**F**
TBM — electrics				**F**	**F**
Diesel plant			**F**		**F**
Battery locomotives				**F**	**F**
Fuel store			**P**		**P**
Battery charging					**P**
Compressed-air workings	**F**	**F**	**P**		
Timber headings, break-outs, etc.	**F**		**P**		
F = Fixed **P = Portable**					

Table 6 — Portable fire extinguishing equipment

Class of materials involved	Extinguishing medium
Fires involving solid usually of an organic nature, in which combustion normally takes place with the formation of glowing embers	Water extinguisher
Fires involving liquids or liquefiable solids	Foam extinguisher, CO2, dry powder
Fires involving gases	Water spray to cool cylinder, foam to extinguish any fire when valve has been closed
Fires involving metals	Dry powder, dry sand Electrical equipment (if live)
Electrical equipment (if live)	Inert gas, dry powder, dry sand

56.3.2In any place where the use of timber would introduce a special hazard, whether by reason of the location's vulnerability, or because the consequences of fire are particularly serious, steel should be substituted if possible. If straw is stored for emergency stuffing at the face, it should preferably be kept in bales in a damp condition and stored in a metal container. However, it is strongly recommended that straw not be used. Substitutes such as rockwool should be used wherever possible.

56.4 Flammable liquids

56.4.1Flammable liquids shall always be contained in tightly sealed and properly labelled metal containers, which should be stored in a cupboard or bin that is fire-resistant when tested in accordance with BS 476-20, BS 476-21 and BS 476-22, such a bin or cupboard must be locked when materials are not in use. Such liquids shall be stored apart from other combustible materials and at safe distances from areas of high activity, electrical installations and explosive magazines. Not more than one day's supply of such liquids shall be kept in a tunnel.

56.4.2Drip pans/bunds must be provided to catch any leakage from containers.

56.5 Compressed gases

56.5.1 Below ground, cylinders containing oxygen must be segregated from cylinders of flammable gas (e.g. acetylene, propane and butane), except when in use. Under no circumstances shall oxygen cylinders be allowed to come into contact with any form of grease. All cylinders of compressed gases must be kept away from flammable liquids and combustible materials. The cylinders should be of the smallest size practicable for the operation to be undertaken, and shall remain underground only for the period of the operation.

56.5.2 All cylinder valves, including the screw thread into the neck of the cylinder, shall be regularly checked for leaks. Any cylinders found to be leaking must be removed to open air immediately. Gas cylinders are liable to damage from mechanical impact, which could cause leakage from valves or cylinder rupture, leading to explosion. Cylinders should therefore be protected by suitable containers during handling, storage and use, and protected against any risk of impact by falling or being struck by other plant or equipment. Cylinder valves and a fire extinguisher should be immediately available.

56.6 Handheld blowpipes

56.6.1Handheld blowpipes must conform to BS EN ISO 5172 and shall be fitted with non-return valves at both gas inlets, and with flame arrestors with cut-off valves at the gas supply outlets. All safety devices must conform to BS EN 730. Hoses shall conform to BS EN 559, and pressure regulators to BS EN ISO 2503. Transportable acetylene containers must conform to BS EN 1800. Containers for other fuel gases shall conform to BS EN 12862 or the appropriate standard in the BS 5045 series.

56.7 Lighting fixtures

56.7.1Any lighting fixtures in storage areas for combustible materials or flammable liquids shall be constructed, located and maintained so that combustible materials cannot be ignited by the heat that the fixtures produce.

56.8 Accumulation of refuse

56.8.1All combustible refuse shall be removed from a tunnel as frequently as practicable, at least once per shift, to prevent the build-up of a fire hazard however It is preferable that such material be removed in the normal process of working. Combustible refuse must not be deposited close to any other combustible materials. Any combustible refuse that cannot be removed shall be stored in metal bins with close-fitting lids, and away from naked lights and other fire hazards and ignition sources. These refuse storage areas should have suitable fire-fighting equipment nearby.

56.9 Welding and cutting (burning)

56.10 General

56.10.1Whenever possible, such work should be undertaken at the surface or cold processes should be adopted. In pressurized workings, the risk of fire is significantly increased, as all combustible materials ignite more easily and burn more fiercely, making fire-fighting more difficult.

56.10.2In tunnels where explosive gases are present, hot work shall be prohibited. If there is potential for these gases to be present, thorough atmospheric monitoring must be carried out before work commences and continued throughout the work period. In the event of an explosive atmosphere being detected, all hot work must cease immediately.

56.10.3A "permit to work" system shall apply to any underground welding or cutting. This should specify the conditions for storage, transport and use of equipment, and the fire precautions. The "permit to work" system must also cover the return of the equipment

56.10.4To surface storage. "Permits to work" shall have specific dates, as an open-ended-permits are not considered acceptable. No work shall be carried out on fuel or oil tanks until they have been purged and certified gas-free.

56.10.5On completion of the operation, the area shall be inspected to check that nothing is smouldering. A fire-watcher with fire hose or extinguisher shall be in attendance during the operation and for at least 1 h after its completion.

56.10.6No person carrying out welding or cutting shall be allowed to work alone, as they could fail to notice a fire developing or be unable to extinguish it successfully. However, all unnecessary personnel shall be excluded from the work area. Fumes are generated by many welding and cutting operations, and so good ventilation is important. When worked, galvanized steel produces toxic fumes that must not be inhaled. Local exhaust ventilation, airlines and air movers are particularly effective in dispersing local concentrations of fumes.

56.11 Electric arc welding and cutting

56.11.1Petrol-driven generator sets shall not be permitted underground. Electrically-powered transformer or rotary converter sets must be used as the power supplies for the generation of electric arcs. If diesel sets are used, adequate ventilation is essential, and additional fire precautions must be taken and proper arrangements made for the storage and handling of fuel . Carbon electrodes may be used for arc-air gouging provided that there is adequate ventilation. Arc welding plant, equipment, accessories and installations should conform to BS 638-4, BS 638-5 and BS 638-7.

56.12 Fires involving electrical equipment

56.12.1The special hazard of fumes and smoke from burning PVC or similar cables must be taken into account. Because of the restricted space in tunnels and the usually very damp conditions, it is particularly important that arrangements for isolating defective cables or equipment be carefully planned so that persons are not exposed to the hazards of electric shock and electrocution.

56.12.2The network of cables must be planned so that essential fire-fighting resources, including pumps, lighting and ventilation, are not cut off in the process of isolating overheated equipment, and so that signals and communications are maintained. Electrical fittings in areas where gas cylinders are stored shall be protected so that they do not present a possible ignition source.

57. Fire precautions, Protection, Fighting System and Rescue.

57.1 General

57.1.1The advice and that of the local fire service shall be sought before the quantity, type and position of fire protection equipment is decided. In addition to fire-fighting to save life, account must be taken of the consequences of fire damage to the tunnel structure and the hazards associated with any remedial work required.

57.2 Fire mains and hose connections

57.2.1A fire main conforming to BS 5306-1 / relevant IS, providing water for fire-fighting, shall be made available throughout the tunnel, with hydrant outlets at intervals not exceeding 50 m. Such outlets must be clearly marked and be readily accessible. The water supply shall be sufficient in volume and pressure for the operation of fire hoses, water sprays or other fire-fighting equipment, and the advice of the fire service must be sought. Equipment shall be located strategically in accordance with the progress of the tunnel, and must be regularly tested and properly maintained.

57.3 Fire extinguishing systems

57.3.1 All TBMs shall have fixed fire extinguishing systems (Refer clause No :56.3.1)

57.3.2 Portable fire extinguishers conforming to BS 7863 or IS 2190: 1979 Code of Practice for the selection, installation and maintenance of portable fire extinguishers and to the appropriate part(s) of BS EN 3-1 shall be provided. They shall be selected and installed in accordance with BS 5306-8-2000 and maintained in accordance with BS 5306-3-2003. They must be sited so that they are readily accessible by personnel. The effects of the extinguishing medium on the tunnel atmosphere must also be taken into account in the selection process. Table 5 and Table 6 indicate suitable extinguishing media for a range of fire locations. BS 5306-1-2006 gives guidance on the selection of appropriate systems. Recommendations concerning the application of fixed or portable installations are given in Table 5.

57.3.3Distance of travel to Fire extinguisher not exceeding 15 M

57.4 Routine testing and maintenance of fire protection equipment

57.4.1All equipment should be maintained in good working order, and this should be verified by routine testing in accordance with the manufacturer's instructions. Records of such testing and inspection shall be made available to the Employer / Engineer upon request.

57.5 Vulnerable items and locations

57.5.1Certain items and locations are particularly vulnerable to fire, whether by reason of high flammability, or because a fire would be especially difficult to extinguish, or because the consequences of a fire could be catastrophic. These items and locations should be identified, assessed and protected. They include:

a) Timber in headings and face;

b) Stores or accumulations of:

— Timber and scrap timber;

— Paper cement bags;

— Straw;

— Oily rags and hemp;

— Flammable liquids and gases;

c) Conveyor belts;

d) Cables and wiring;

e) Electrical installations, including transformers and switchgear;

f) Hydraulic (oil) systems;

g) Flame cutting and welding areas;

h) All areas of compressed-air working;

i) All methane-bearing strata;

j) Storage areas for explosives;

k) Shafts and all openings to the tunnel;

l) Coal seams liable to spontaneous combustion.

57.6 Escape routes

57.6.1All escape routes underground shall be clearly signed. Blind headings must be marked as such. Consideration shall be given to the provision of places of relative safety/refuges, where appropriate however the suitability and location is subject to a full risk assessment and notice of no objection from the Employer / Engineer.

57.7 Fire-fighting and rescue

57.7.1In addition to the hazard of fire, rescue of persons injured by falls and other accidents should be considered; this is especially important in the case of a tunnelling machine that uses compressed air in the cutter-head chamber. Lock-keepers and workers should be trained in the routine for rescuing a person injured in the working chamber (the cutter-head chamber). They should also be familiar with the manufacturer's instructions for operation of the airlock(s) on the machine.

NOTE When there is a loss of pressure in the cutter-head chamber there could be a problem in closing the door from there to the airlock in order to recompress the persons in the airlock.

57.7.2When a shaft is to be sunk using a vertical airlock, the method of removing an injured person from the working chamber should be given consideration. The normal way to enter such an airlock from the chamber is by climbing a ladder through a hatch and into a small chamber in which it is only convenient to stand. An alternative means of egress for an injured person should therefore be provided.

NOTE The most common way is to provide a man-riding cage that can be lowered through a muck lock, into which the injured person can be lowered in a special emergency sling. The man-riding cage can be covered with a hood or mesh so that no part of the person being rescued can be trapped in the guides and doors. Access to the airlock should be tested to verify that it is adequate for persons wearing breathing apparatus.

57.8 Fire services and operational capacity

57.8.1During the planning stages of construction the cooperation of the fire service should be sought in making arrangements for fire-fighting and rescue and for summoning of the fire service.

57.8.2An assessment must be carried out during the planning stage of the tunnel project, in consultation with the fire service, to establish their operational capacity including pre-determined attendance times and to establish details of any additional facilities and equipment required for operations beyond that capacity. A practical test of the plan shall be carried out in liaison with the fire services.

57.8.3Because of the extended distances that occur in tunnel construction, special arrangements and facilities may be necessary to assist fire services in dealing with incidents occurring in tunnels during the construction stage. Fire service operational capacity in tunnels is based on the principle that fire service personnel are able to take equipment to the scene of an incident, deal with the incident and return to ground level safely. Equipment and other issues that should be considered include some or all of the following:

a) provision of a suitable control room and/or a bridgehead location(s) from which resources and operations can be controlled close to the incident;
b) underground transportation of personnel and equipment to the incident or bridgehead;
c) lighting, communications, ventilation, and smoke control;
d) fixed installations, e.g. water mains, fire suppression systems, fire-fighting hose;
e) portable fire-fighting equipment;
f) provision of extended-duration breathing apparatus;
g) training, and site familiarization for fire service personnel.

57.8.4 Following an assessment of operational capacity, the fire service may be unable to provide an assurance that their personnel will be able to deal with all incidents within the tunnel. In this event, the fire service must be asked to formally notify the contractor and give details of their reduced operational capacity for the particular tunnel or stage of the tunnel construction. Thereafter, the contractor should make their own arrangements for fire-fighting and rescue. The results of this assessment must be issued to the Employer / Engineer who will verify that the contractor has taken the appropriate measures as advised by the fire service.

57.9 Emergency control facilities

57.10 Control rooms

57.10.1An emergency control room shall be provided, from which the senior officers of the emergency services can control their response to an incident. Such a room shall be equipped with ex-directory telephone lines, site radio communications and drawings showing the up-to-date layout of the underground workings.

57.10.2Where practicable, an emergency control room should be provided at the top of each shaft or other point of access. If this is not possible, an up-to-date weatherproof drawing showing the depths of shafts, layout of tunnels and location of fire-fighting equipment shall be displayed at these locations. It must also give details of where and how to notify the fire service.

57.10.3On multiple-contractor projects, or once multiple access points or exits are established, control rooms must be interlinked. In larger tunnels, and where requested by the fire service, a “leaky feeder” communication system compatible with the fire service’s radio system shall be provided for use underground.

57.10.4Full particulars of the ventilation system in use, including details of all fans and ducts, and of any apertures and doors, shall be kept available for use by the fire service officers or other responsible persons. A suitably qualified member of the contractor’s staff must also be made available to the fire service to operate the ventilation system should this be necessary. Details of all “permits to work” relating to the use of hot cutting/burning equipment must be available to the fire service and in the emergency control room as this will provide and indication of works being undertaken and the likelihood of hazardous/explosive substances that may be in the vicinity of the incident/fire.

57.11 **Bridgeheads**

57.11.1A forward control point, or “bridgehead”, shall be provided in situations where it may be necessary for breathing apparatus or other operations to be started up at a distance from the original point(s) of entry to a hazardous area, whilst remaining in a safe-air environment.

57.12 Raising the alarm

57.12.1Adequate arrangements shall be made for raising the alarm and calling the fire service in the event of a fire underground. The fire alarm system shall be extended and modified as the tunnelling work proceeds. The alarm should be clearly perceptible to all persons in the workings, and to key personnel above ground. Warning arrangements can include any or a combination of the following, as appropriate:

— Vocal warnings;
— Telephone communications;
— Hand-operated or electrically-operated bells or sirens;
— Special flashing lights;
— Flashing of the main lighting circuits.

57.12.2 The locations of the fire alarm devices shall take into account the layout of the working areas underground, the adverse nature of the working conditions (especially high noise levels) and the transient nature of the working locations. Where single alarm systems are used, wiring must conform to BS 6387:1994, meeting a minimum category rating of AWZ [i.e. the lowest performance category for resistance to fire (A), resistance to fire with water (W), and the highest performance category for resistance to fire with mechanical shock (Z)].

57.12.3 A separate and distinct signal to order the total evacuation of the workings must be provided. The fire alarm systems should be regularly tested and properly maintained. Records of such test shall be regularly inspected by the Employer / Engineer.

57.12.4The locations of the alarm devices should take into account the layout of the working areas underground, the adverse nature of the working conditions (especially high noise levels) and the transient nature of the working locations

57.13 Alarm procedures

57.14 Consultation and planning

57.14.1Specific procedures should be prearranged for each site. The contractor shall consult the fire service for advice on fire and evacuation procedures. Planning must cover matters such as arrangements for fire-fighting, rescue and evacuation, and the management controls necessary to affect this. Protocols for calling the fire service shall include instructions to switchboard/security staff on how emergency calls are to be made, and the information that is to be passed to the fire service control.

57.15 Action at point of discovery of fire or smoke

57.15.1All personnel should be given the following set of instructions to follow on the spot when there is a fire or suspicion of fire in a tunnel or underground working.

a) Raise the alarm in the tunnel and attack the fire only if it is safe to do so.
b) Report the fire, stating the following;
   — Where the fire is;
   — What is on fire;
   — Whether evacuation is in progress.

57.16 Action on receiving the alarm

57.16.1The person on site receiving the alarm must immediately undertake the following:

a) Summon the fire service, giving a precise rendezvous point, and arrange for a responsible person to meet them on arrival;
b) Inform the senior site manager or emergency coordinator and the Employer / Engineer.

57.17 Site training

57.17.1All underground personnel shall receive training in the care and use of fire-fighting equipment. All site personnel shall be made familiar with the fire emergency procedures prescribed. Fire drills must be held to familiarize all personnel with the practical working of the systems.

57.18 Access

57.18.1Access to the site, and the provision of, and access to, hard standing for fire appliances and ambulances is of vital importance and must be included in the site emergency procedures, following consultation with the local fire service and ambulance services. Access to the shaft and workings must be available for the emergency services at all times. The surface of any walkways in the tunnel shall be maintained in a safe condition so that they can be used safely by fire-fighters, even in conditions of nil visibility.

57.19 Lighting

57.19.1Emergency lighting shall be provided and maintained at all times, particularly at fire points, escape routes, emergency exits and tunnel access points.

57.20 Smoke control

57.20.1Smoke is a major danger in fire, as it can cause asphyxiation. It can interfere with visibility, resulting in disorientation and possibly panic. The hazard can be reduced by using the ventilation system to the best advantage. The problem shall be studied in advance in liaison with the fire service. The aim should be to configure the airflow so that it clears the heat and smoke without spreading the fire.

57.20.2 Valved discharge points can be provided at intervals on an air line along a tunnel to provide air for persons trapped by fire and smoke. Self-rescuers and breathing apparatus must also be provided, as these are necessary in severe conditions of smoke. The use of a water spray curtain at the outbye end of a TBM will significantly control the spread of smoke from a fire on the machine.

57.21 Rescue facilities

57.21.1 First aid equipment, including stretchers, shall be available. Rescue equipment, including full breathing apparatus (SCBA/CCBA), shall be provided and maintained so that it is readily available in an emergency. All rescue equipment must be stored in containers designed to provide protection from adverse conditions likely to be encountered in the workings. In many tunnels, a dedicated rescue skip or train will be necessary for quick response by a site rescue team and/or emergency services, the provision of such equipment shall be determined through risk assessment.

57.22 Self-rescuers

57.22.1 All persons underground should have immediate access to a self-rescuer which provides the user with a supply of oxygen for at least 60min whilst walking. Where such oxygen supplies are likely to be needed for longer periods, stockpiles of additional rescuers shall be provided at intervals, e.g. in long tunnels. Self-rescuers should normally be worn on a belt by persons underground, but may be stored in racks that are immediately accessible for use in an emergency. Where alternative escape routes are available, racks of self-rescuers shall be provided for each escape route. If self-rescuers are stored on racks, there must be a maintenance and inspection system to check that a sufficient number are always available and that they are always in full working order.

57.22.2 The highest standard of cleanliness must be maintained in situations where oxygen is used. Specialist advice should be sought on how this is best achieved. For guidance on the use of self-rescuers in compressed air.

58. Access

58.1 Walkways

58.1.1 Pedestrian access is generally provided by walkways. Every tunnel shall have a safe pedestrian route from face to pit bottom and then by stair tower to the surface. In shafts and tunnels where man-riding is carried out, a walkway may be required only for maintenance access or for emergency evacuation.

58.1.2 The pedestrian route shall be clearly marked. A walkway shall provide a minimum clear space of 2000 mm high and 900 mm wide with a walking surface not less than 430 mm wide. Where a walkway is adjacent to a traffic route, precautions must be taken to safeguard any persons using it during the passage of vehicles, unless the simultaneous passage of pedestrians and vehicles is prevented.

58.1.3 Extra clearances must be allowed for side-throw, end-throw and sway of vehicles and for the contingency of derailment. Where continuous clearance with the recommended minimum dimensions is not practicable, refuges must be provided at intervals related to the curvature and gradient of the tunnel and to traffic speed.

58.1.4The visibility of both vehicle drivers and pedestrians can be adversely affected by dust or humidity in the atmosphere. Such refuges shall be clearly illuminated and may comprise:

a) Recesses constructed outside the general line of excavation;
b) Robust barriers between the safe position and the haulage route;
c) Stages raised clear of any possible hazard.

58.1.5Refuges must also be provided where there is no space for a segregated pedestrian walkway and man-riding is not in use. In addition, a safe system of work shall be implemented to protect persons who are required to be in the tunnel from the danger of tunnel transport. The walkway surface must be maintained in good condition and shall incorporate adequate grip and be free from slipping, tripping or falling hazards such as irregularities, sudden changes in level, loose boards, obstructions, etc. It could be required by the emergency services in conditions of nil visibility.

58.1.6Elevated walkways from which a fall of 1.5 m or more is possible shall be furnished with suitable guard rails and toe boards. Hand rails or similar shall be provided on inclined walkways. All stairways shall be provided with guard rails and toe boards. Water should not be allowed to stand or accumulate at or above walkway level at any point. Any accidental spillage of muck or other material on a walkway must be cleared away promptly.

58.1.7It is important to provide adequate lighting along a walkway, not only to assist pedestrians, but also to allow haulage drivers to see pedestrians clearly. The lights should be located so that they are not obscured by passing vehicles. In unlit tunnels, where only occasional access is required for inspection and maintenance, refuges shall be identified by reflective material. As necessary, reflective signs indicating escape routes shall also be posted. Persons entering unlit tunnels must be equipped with a suitable hand/cap lamp and appropriate means of communication.

58.1.8Access on foot shall not be permitted in a tunnel whilst a winch haulage is in operation, unless there is adequate protection to the walkway.

58.2 Access for maintenance

58.2.1Where access in tunnels is required for maintenance/minor repair purposes only, the risk assessment should make provision for this. Any safe system of work devised should address:

a) Minimum dimensions for access, which are set out in prEN 12336 .
b) Adequate ventilation;
c) Gas monitors and self-rescue sets;
d) Rescue procedures, with due allowance made for limited access and any reduced number of Personnel on site;
e) The type of breathing apparatus and stretcher available;
f) The need for at least two persons in any one location.

58.3 Working at height in tunnels and shafts

58.3.1Where work at height is required in tunnels and shafts, the system used must be fit for purpose. For work such as the installation of bolts in tunnel linings or mesh in sprayed concrete linings, mechanical access equipment shall normally be used. Fixed ladders and access platforms or permanently installed moveable platforms shall be provided where work at height is regularly required at the same location, e.g. to place and tighten tunnel lining bolts on TBMs. For short duration work at height, ladders are acceptable provided that they can be properly grounded and secured. Mobile access towers shall not normally be used for work at height, given the rigours of the tunnel environment.

59. Blasting Operations

59.1 General

59.1.1The Contractor shall design and implement his blasting techniques so as to minimise dust, noise, vibration generation and prevention fly rock.

59.1.2Blasting technique should be consistent not only with nature and quaintly of rock to be blasted but also the location of blasting.

59.2 Authorisation for Blasting

59.2.1The Contractor shall ensure that all blasting operations will only be permitted following consultations with the relevant authorities and subsequent issuing of the permission to blast permits. The Employer / Engineer must also give his consent in writing before any blasting operations take place.

59.2.2All blasting shall be conducted under the direct supervision of a Licensed Shotfirer.

59.3 Risk Assessment and Method Statements

59.3.1The Contractor shall produce a detailed hazard and risk assessment and an in depth method statement for amongst others the following elements:

(a) Type of explosives to be used.
(b) Anticipated effects of vibration on nearby structures.
(c) Blasting patterns.
(d) Delivery of the explosives.
(e) Transportation and storage of explosives on site.
(f) Drilling and charging of holes.
(g) Warning sirens.
(h) Measurement of Vibration
(i) Provision of sentries.
(j) Use of blast screens.
(k) ALL CLEAR.
(l) Ventilation following blasting.
(m) Atmosphere monitoring.
(n) Procedure for miss-fires.

60. Drilling

60.1 Drilling General

60.1.1Appropriate precautions should be taken before drilling commences including barring and scaling the tunnel face and crown adjacent to the face, clearing the face and examining it and the muck pile for misfires. These should be dealt with in accordance with appropriate regulations.

60.1.2Mark round on tunnel face with material which can easily be seen and not removed, e.g. paint. Avoid drilling into existing sockets.

60.1.3Accurate marking, drilling and loading of the round will minimise shattering of surrounding rock.

61. Compressed air works

61.1 General Measures

61.1.1Identification badges Should be given to all employees, indicating that the wearer is a compressed air Worker. A permanent record should be kept of all identification badges issued. The badge which should be worn at all times off the job as well as on the job should contain the following information;

i) Employees name

ii) Address of medical lock; and

iii) The telephone number of the appointed medical practitioner.

61.1.2 It should also contain instructions that in case of sudden illness or emergency of unknown or doubtful cause the wearer should be rushed to the medical lock.

61.1.3 When no lock attendant is inside a man lock or working chamber the arrangements should be such as to enable any person employed in compressed air inside the man lock or working Chamber to control the door of the man lock or warning chamber in order to leave or enter the working chamber in the case of emergency.

61.1.4 No person should consume alcohol whilst he is employed in compressed air.

61.1.5 No person under the influence of alcohol should undergo compression in any lock other than India medical lock.

61.1.6 No person should smoke whilst he is employed in compressed air.

61.1.7 No person should wilfully

i) Obstruct; or

ii) Delay; or

iii) Refuse to follow any instruction given by a lock attendant or medical lock attendant in the course of his duties.

61.2 Medical Examination

61.2.1To be able to enter a compressed air environment the worker must present proof of hyperbaric medical fitness.

61.2.2Personnel shall, except under exceptional circumstances, not be younger than 18.

61.2.3A comprehensive medical examination prior to entering a compressed air environment for the first time is required along with an annual comprehensive medical examination and periodic medical checks thereafter.

61.2.4 Workers suffering from the following ailments should not enter a compressed air environment.

a) chronic infection of the nasal ducts (sinusitis)
b) chronic ostitis, perforated eardrum, permanent obstruction of the Eustachian tube
c) chronic lung infection (bronchitis, asthma, tuberculosis, etc.)

d) heart infections
e) epilepsy
f) serious loss of hearing and sight
g) diabetes
h) psychiatric deviations
i) alcoholism
j) ulcers
k) hernia
l) anaemia
m) chronic kidney infections
n) some skin infections.

61.2.5 The medical examination and checks should be undertaken by a doctor competent in hyperbaric medicine. The doctor should advise on the content of the examinations but as a minimum, the following examinations but as a minimum, the following examinations should be made:

a) electrocardiogram including one under physical stress – annually.
b) ear, nose and throat – annually.
c) spirometry (functioning of the kings) – annually.
d) blood tests – complete blood count- annually.
e) pressure test in decompression tank or room.
f) lung and long bone x-rays (at the doctor's discretion)

61.3 Competent persons for a supply plant

61.3.1The contractor should appoint your suitably qualified or experienced person to be in charge of the art supply plant at a construction site.

61.3.2A person appointed at all times be: -

i) In charge of the plant; and

ii) In attendance at the construction site when any person is employed in compressed air at the construction site

iii) There should be also an experienced, competent and reliable person on duty at the, air control valves as a gauge tender who will regulate the pressure in the working areas

61.4 Air-Supply Plant

61.4.1The air supply plant for the production and supply of compressed air to any man lock working chamber and medical lock should be of suitable design.

61.4.2The air intakes for all air compressors should be located as far as practicable at a place where fumes exhaust gases and other air contaminants will be at a minimum. Gauges indicating the pressure in the working chamber should be installed in the compressor building the lock attendant station and at the contractors site office

61.5 Low Air Compressor

61.5.1 The Low air compressor plant should be of sufficient capacity to ensure that the work is carried out safely

61.5.2 Air compressor units should have at least two independent and separate sources of power supply and each should be capable of operating the entire low air plant and its ancillary systems

61.5.3 Switching from one independent source of power supply to the other should be done periodically to ensure the workability of the apparatus in an emergency.

61.5.4 The air mains supplying the working chamber and airlocks should be properly protected and where failure could result in danger to the workings should be duplicated.

61.6 Man Locks

61.6.1 Every man-lock should be of adequate internal dimensions and capacity for the purposes for which it is used

61.6.2 Every man-lock should have an independent supply of fresh air which can be used for ventilation

61.6.3 Every man-lock should be provided with suitable equipment including –

i) Valves are taps controlling the flow of air into and from the man-lock so as to enable careful compression and accurate decompression to be carried out;

ii) A clock or clocks so positioned that the lock attendant and any person in the man-lock can readily ascertain the time;

iii) Pressure Gauges which will readily indicate –

a) To the Lock attendant the pressure in the Man-lock and in each working chamber to which the man-lock affords direct or indirect access;

b) To persons in the man-lock the pressure in the Man-lock; and

61.6.4 A notice which can be easily read and understood by the workers, indicating,

i) The precautions to be taken by persons during the compression and decompression and after decompression;

ii) The maximum number of persons who may normally be accommodate man-lock.

iii) Should be fixed in each man-lock.

iv) All furniture and equipment in a man-lock Should be incombustible or of fireproof material

v) When the pressure in a working chamber exceeds 1 bar, the man-lock should be used solely for the Compression and decompression of persons.

vi) No person should be employed in compressed air unless – He has had previous experience of such work; or If he has not had such experience, He is accompanied by a person experienced in such work.

61.7 Compression and decompression

61.7.1Compression of all persons in the man-lock should be carried out according to the compression procedures

61.7.2Decompression of all persons to normal conditions should be in accordance with their. decompression procedure and should be carried out under supervision of the appointment medical practitioner

61.7.3Where a person in man-lock collapses or is taken ill during decompression, the lock attendant in-charge should raise the pressure in the working chamber and should report the matter immediately to the medical lock attendant on duty the lock attendant should then follow the instruction of the medical lock attendant or the appointed medical practitioner on duty

61.8 Medical Locks

61.8.1Where persons are employed in compressed air in a working chamber, a suitably constructed medical lock should be provided and maintained.

61.8.2Where more than 10 persons so employed medical lock should be provided.

61.8.3Every medical lock should be situated as near as possible to the Man-lock.

61.8.4Every medical lock should –

i)Have not less than 1.8 m clear headroom at its height point;

ii)Have 2 compartments so that it can be entered while under pressure.

iii)How efficient means of verbal communication and means of giving nonverbal signals between persons inside and outside the medical lock and between persons in compartments of the medical lock;

iv)Have one or more Windows through which any person in either compartment of the medical lock can be observed from the outside;

v)Be adequately ventilated;

vi)Be protected from the weather and the sun;

vii)Have adequate lighting.

61.8.5Every medical lock should be provided with suitable equipment including: -

i)A pressure recording gauge which should be accurate within 0.05 bar;

ii)If a circular recording chart is used it should rotate at a speed of not less than once in 4 hours;

iii)a couch not less than 1.8 metres length.

61.8.6Medical lock should not be used for any purpose other than a therapeutic purpose, but it may be used:

i) For training and testing of persons without previous experience of compressed air work; or

II) In case of emergency.

iii) A medical lock should at all times be kept ready for immediate use.

iv) External lighting should be used however if internal wiring is necessary then such wiring should be of mineral insulated copper cable protected by stainless steel tubing complying with British Standard 6207 or equivalent standards.

v) All spark or arc-creating devices such as switches, standard light bulbs, and standard electrical outlets should not be used within a medical lock.

vi) A medical lock is an air receiver. It should be tested and registered by department of occupational safety and health.

vii) A Medical lock should be capable of withstanding a pressure of not less than 100 Kn/m^2 above the maximum pressure used or likely to be used in the chamber at any time.

61.9 Man-lock and medical lock

61.9.1Every man-lock or medical lock should be under the charge of a lock attendant or medical lock attendant respectively.

61.9.2That should be not less than 3 man-lock or medical lock attendants available for duty in respect of each man-lock or medical lock attendant respectively.

61.9.3No man-lock or medical lock attendant should be on duty for more than 12 hours in any 1 shift.

61.9.4All man lock and medical lock attendance should:

i) Be medically fit;

ii) Be willing to go into compressed air at any working pressure;

iii) Be trained in first aid; and

iv) Have completed a suitable training course designed to familiarise them with the problems associated with compression decompression and compressed air illness and with the keeping of records

61.9.5The Man-lock attendant should attend the man-lock or in a working chamber to which the man-lock affords direct or indirect access.

61.9.6The medical lock attendant should attend the medical lock: -

i)Whilst any person is employed in compressed air at a pressure exceeding 1 bar.

ii)Whilst any person is being treated in the medical lock; and iii)For 24 hours after the last man-lock decompression from a pressure exceeding 1 bar has taken place.

61.9.7 A medical lock attendant should when on duty have immediate access to all records of precious and other relevant information regarding conditions in the man-lock and working chamber.

61.10 Duties of man-lock attendants

61.10.1The Lock attendant in charge of a man-lock should: -

i)Control the compression of all persons in the man-lock

ii)Carry out the compression of all such persons according to the compression procedure

iii)Control the decompression of all persons who have been employed in compressed air; and;

iv)When a person has been employed in compressed air at a pressure exceeding 1 bar carry out the decompression of the person according to year decompression procedure set up appointed medical practitioner.

61.10.2Where any person is employed in compressed air in a working chamber the lock attendant in charge should: -

i) Record in the lock attendant's register and any man-lock decompression chart and give them to the medical practitioner.

ii) Keep in his custody the lock attendant's register and any man-lock decompression chart and give them to the medical practitioner.

61.10.3 The contractor should ensure that the lock attendant in charge of a man- lock complies with British Standard 6207 or equivalent standards.

61.11 Appointment of Medical Practitioners

61.11.1Where any construction work in compressed air is carried out, the contractor should appoint a medical practitioner to supervise all medical matters which may arise in connection with the construction work.

61.11.2The appointed medical practitioner should be: -

i) Suitably qualified and conversant with the problems associated with work in compressed air and the medical aspects of such work;

ii) Conversant with the provisions of these guidelines.

iii) Medically fit;

iv) Willing go in to compressed ait at any working pressure; and

v) On call at all times when construction work in compressed air is in progress.

61.11.3The appointed medical practitioner should: -

i) Examine medically persons for fitness for employment in compressed air;

ii) Supervise medical lock attendants;

iii) Treat compressed air lines and other conditions; and

iv) Supervise the keeping of: -

v) All compressed air workers medical information, including the Compressed Air Workers Medical Records.

vi) The compressed air workers' Compressed Air Illness Notification Form.

vi) The compressed air workers' individual records and

vii) Ensure that all the above records are readily available for submission to the Employer / Engineer when required.

61.12 Medical Supervision and Certification

61.12.1No person should be employed in compressed air unless he has been examined by an appointed medical practitioner and certified by hi, to be fit for such employment.

61.12.2 In the case of a person who: -

a) Is a new starter, the examination for certification of fitness should be carried out not more than 3 days prior to work in the compressed air;

b) Is not a new starter and who continues to be employed in compressed air at a working pressure not exceeding 1 bar, the examination for certification of fitness should be carried out not less than once in every 3 months;

c) Is not a new starter and who continues to be employed in compressed air at a working pressure exceeding 1 bar, the examination for certification of fitness should be carried out not less than once in every 4 weeks.

61.12.3A person who is required to be employed in compressed air and who is suffering from a cold in the head, chest infection, sore throat or earache should report the matter to the appointed medical practitioner should recommend to the Chief Safety Manager whether he is fit to work or not.

61.12.4Any person suffering from a cold in the head, chest infection, sore throat, earache or any other illness or injury necessitating absence from work for more than 3 consecutive days should be re-examined by an appointed medical practitioner and certified by him to be fir for employment in compressed air before he resumes such employment.

61.13 Radiographic Examinations

61.13.1Every person employed in compressed air at a pressure exceeding 1 bar should undergo a radiographic examination of his major joints within 4 weeks after starting such employment, unless the person has such an examination during the 12 months immediately preceding such employment.

61.13.2Every person who continues to be employed in compressed air at a pressure exceeding 1 bar should undergo such an examination at intervals of about but not more than 12 months.

61.13.3These radiographic examinations should be arranged by the appointed medical practitioner should enter the particulars of such radiographic examinations in the Compressed Air Worker's Medical Record

61.13.4Every person employed a proposed to be employed in compressed air should submit himself for the necessary medical examinations.

61.14 Working chambers:

61.14.1All parts of the working chamber should be so designed that they can withstand the specified pressure. The effects of this pressure on the surrounding ground should be considered.

61.14.2The electrical installations should be unaffected by pressure and be waterproof, dustproof and explosion protected.

61.14.2An emergency power system via fire hardened cables should be provided to power safety critical equipment. In the event of power failure, the emergency lighting should switch on automatically. The power source for the automatic emergency lighting system should be above ground.

61.14.4At the working face there should be adequate fixed lighting.

61.14.5The air temperature in the workings should be maintained between 15^{0}C and 30^{0}C

61.16 Bulkheads and Air Locks

61.16.1All Bulkheads and air-tight diaphragms retaining compressed air within a tunnel or shaft should be capable of carrying the full thrust of the compressed air at its maximum pressure.

61.16.2If the bulkhead is across a tunnel, its anchorage into the tunnel walls should be fully adequate to support the total load and any air lock or other sealed aperture should be properly anchored into the bulkhead.

61.16.3Tests should be conducted on airlocks at working pressure before being put to use

61.16.4Bulkheads should be tested in situ at working pressure

61.16.5If the diaphragm takes the form of a horizontal air deck across a shaft, it should anchor sufficient mass to resist the thrusts by gravitational force alone and should not rely on the tensile strength of cast iron or concrete

61.16.6The deck should be designed to take all the dead loads when the shaft is under pressure.

61.16.7Air lock Construction should conform with the following: -

i) Should have adequate strength to with stand any air pressure internal or external to which the structure may be subjected in use and in an emergency.

ii) Lock doors are to be of Steel

iii) Should be of adequate dimensions for the full number of men likely to use the lock at any time.

iv) Anchorage of the lock against the thrusts imposed by air pressures on the ends of the lock should be designed to carry all loads safely.

v) Air tightness for the lock itself and satisfactory devices should be provided for ceiling the doors even at low pressures the inner walls of the locks should be suitably treated to eliminate the escape of air; and

vi) Incombustible materials to be used as extensively as possible

61.17 Firefighting equipment:

61.17.1Water jets and sprays should be the principal means of extinguishing a fire in compressed air.

61.17.2As normally pressurized fire extinguishers cannot be relied on to function effectively under pressure, hyperbaric extinguishers should be provided.

61.17.3Breathing apparatus for use in smoke and fumes should only be employed by those trained in its use. Only self-contained compressed air breathing apparatus, (of the type that does not have an air cushion seal to the mask) should be used. The effects of working in a pressurized environment on the duration of the apparatus should be taken into account.

61.17.4Firefighting in compressed air calls for special training and a site fire squad should be designated and trained for firefighting. Firefighting should only be attempted to save life.

61.17.5Methane gas in coal seams can occur at high pressures and could form an explosive mixture with the compressed air in the working chamber. Any possibility of the presence of methane therefore calls for careful study, detection, assessment and additional safety precautions.

61.18 Geological Controls

61.18.1During the construction of the works the geological data must be continuously compared with what is actually encountered. If necessary the construction methods may have to be adapted to suit changing conditions.

61.18.2Allowance should be made for variations in ground water level due to the tides, the presence of water under pressure etc.

61.18.3To minimize risk from working in compressed air, the lowest possible air pressure should be used at all times. To this end it can be beneficial to initially lower the groundwater level. The effect of this on the surrounding ground should be thoroughly investigated. The compressors should be able to counteract the full water pressure in an emergency.

61.18.4In day the pressure of pre-consolidated layers could cause large distortions. By adjusting the air pressure the distortions can be limited but not eliminated and grouting may by necessary.

61.18.5In highly porous ground (e.g. coarse sand) air loss can make it impossible to achieve the required working pressure. Air loss can be reduced by covering the face in an impermeable material such as plastic sheeting or day by injection grouting.

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61.18.6Materials used for injection grouting should not give off toxic gases.

61.18.7Consideration should be given to the possible presence of obstacles like boulders or obstacles near the surface around which compressed air can escape (e.g. old sewers, shafts, boreholes etc.)

61.18.8The importance of sufficient ground cover above the works to prevent ‘‘blow-up’’ should be considered.

61.19 Control of The Air in the Working Chamber and the Lock

61.19.1The air in which the workers are working should be as clean as possible.

61.19.2In the working chamber and especially at the face the air quality should be constantly monitored and additional ventilation air supplied as necessary. Allowance should be made for the fact that the air could be contaminated by the nature of the ground (lignite oxidising materials, grouting materials, presence of methane, etc.) and by the equipment being used (e.g. oil mist from pneumatic tools).

61.19.3Monitoring instruments should clearly indicate the buildup of dangerous contaminants in the air (O_2, CO, CO_2, SO_2, NO_X, CH_4). The ventilation system must allow for the size of the working chamber and the lock and also for the various contaminating factors.

61.20 Precautions to be taken after Work

61.20.1Hose working under pressure should be instructed in the risks associated with such work.

61.20.2Decompression illness can occur anytime during the first few hours after leaving the lock. During this period a worker should be able to easily contact the medical lock attendant on site and to reach a treatment chamber.

61.20.3The worker should not participate during the first few hours in diving, flying, mountaineering or any strenuous sport or exercise.

61.20.4Following treatment for decompression illness, the doctor should examine the worker and certify the period out of compressed air which is required before returning to compressed air working. In principle this should be at least 12 hours.

61.21 Maintenance of Records

61.21.1The contractor should maintain a register of all persons employed in compressed air. Such a register should include the following particulars of the person’s:

i) Name

ii) Identity card or work permit number

iii) Date of birth

iv) Nationality

v) Home address and

vi) Occupation

vii) The contractor should make available a copy of all medical records of that particular person employed in compressed air on termination of is employment.

62. Pressure Vessels (Air Receivers)

62.1.1Pressure vessels should be constructed and inspected in accordance with national statutory requirements.

62.1.2Pressure vessels should be kept clean and free from mineral oil or other inflammable material which could ignite under working

62.1.3Pressure vessels should be maintained in a safe working condition at all times.

62.1.4Daily maintenance should include draining of any liquid which may have collected in the receiver, for this purpose a suitable drain controlled by a valve should be provided at the lowest part of the receiver.

62.1.5Every pressure vessel should be inspected and tested before being commissioned and thereafter inspected at regular intervals of not more than five years.

62.1.6They should be hydraulically pressure tested at regular intervals of not more than two years.

62.1.7Every pressure vessel should have a plate securely fixed to it in a conspicuous place bearing
the following particulars:

a) Name of manufacturer
b) Manufacturing number
c) Country of origin
d) Year of manufacture
e) Maximum safe working pressure and temperature
f) Capacity
g) Date of last inspection and test

62.1.8Pressure vessels should not be modified on site.

62.1.9Every pressure vessel should be provided with at least one reliable pressure gauge on which the maximum safe working gauge pressure shall be clearly marked with a red line.

62.1.10Every pressure vessels should be provided with at least one over-pressure relief valve capable of passing the maximum inflow of air to the vessel.

62.1.11To prevent tampering with such valves, they should be kept locked, sealed or otherwise rendered inaccessible to any unauthorized person.

62.1.12Every pressure vessel should be provided with a shut-off valve at the outlet into the mains.

62.2 Air Lines

62.2.1Air lines should be manufactured to a recognized national standard and rated for the expected working pressure.

62.2.2Only pipes and fittings in good condition should be used.

62.2.3Quick coupling metal pipes (including flanged pipes) may be used for the main supply.

62.2.4Reinforced polyethylene or rubber hoses may be used for smaller diameters (less than 200 mm) or for low pressures (up to 7 bar).

62.2.5Reinforced polyethylene or rubber hoses should be fitted with sturdy couplings.

62.2.6Restraints should be fitted across joints in flexible air lines to prevent danger in the event of unexpected disconnection.

62.2.7The air line should not be disconnected unless the supply has been cut off and pressure has been reduced to zero.

62.2.8Air supply lines should be adequately supported both above ground and in shafts and tunnels. They should be protected from mechanical damage.

62.2.9Shut-off valves should be installed in all main supply air lines at regular intervals depending on size of airline but a maximum spacing of 500 m is advisable.

62.2.10Shut-off valves should be fitted at the entrance to the works and near the end of the supply line to close off air supply in case of damage.

62.2.11Discharge valves should be fitted in air lines to reduce air pressure in the air lines when equipment is shut down.

62.2.12Air lines and their fittings should be maintained in good condition and leaks should be repaired promptly both to reduce air loss and control noise emission. A drain should be fitted to an airline to enable moisture present in the air line, to be drained off regularly before entering the equipment.

62.2.13Filters should be installed in the air line near equipment to prevent solids entering the equipment.

63. Use of Explosives

63.1 General

63.1.1Only competent shot firers should be authorised to handle and use explosives. They should be aware of local regulations, if any, and manufacturers recommendations.

63.1.2At change of shifts the shot firer should fully inform his relief of the state of the work at the face.

63.2 Explosives

63.2.1Explosives and detonators not required for immediate use should not be stored in the tunnel.

63.2.2Only water resistant explosives should be used in tunnelling.

63.2.3Under freezing conditions only explosives not susceptible to the cold should be used. In order to ensure adequate transmission of the detonation, cartridge diameter must be more than 30 mm and the space between explosives and surrounding material should be as small as possible.

63.2.4The destruction of old or damaged explosives is a hazardous operation and should be done in accordance with accepted safe practice.

63.2.5Explosives should in all cases, be used in accordance with local or other appropriate regulations.

63.3 Detonators

63.3.1Various types of initiation systems are available and selection should be made with local conditions in mind. Detonators should be stored and transported separately from explosives.

63.4 Firing Sets – Firing Lines

63.4.1The initiator should match the detonators used.

63.4.2Initiators should be adequately maintained.

63.4.3Electrical firing lines should be installed on opposite sides of tunnel from power lines or other conductors such as water lines.

63.4.4All electrical firing lines should be kept short-circuited when not in use.

63.5 Blasting

63.5.1A blasting plan should be drawn up before any blasting is undertaken.

63.5.2All blasting should be carried out in accordance with the plan.

63.6 Transport of Explosives

63.6.1Detonators and other explosives should be transported separately and should be stored apart until used.

63.6.2Only sufficient explosives for immediate use should be transported to the working face.

63.6.3Explosives should be transported in containers and / or vehicles specially designed for the purpose. Such containers and/or vehicles should be dearly marked to distinguish them from other vehicles.

63.6.4A responsible person should always accompany the explosives during transport.

63.7 Charging of the round

63.7.1Charging should not be undertaken until all drilling in that area has been completed.

63.7.2In the event of the threat of thunderstorms, charging with electric detonators should be stopped immediately and the working face evacuated. Non-electric detonators should be used in that case.

63.7.3Charging and blasting should be done by authorised persons.

63.7.4Electric detonators shall only be used where stray currents can be excluded.

63.8 Precautions before blasting

63.8.1Before proceeding with continuity testing of the firing circuit, care should be taken to remove all persons from the danger zone and guards should be placed to prevent access to the danger zone. Persons in working areas adjacent to the blasting site should be warned of pending blasting operations.

63.8.2Guards must remain at their posts until they have received definite instructions to leave.

63.8.3Consideration should be given to the use of non-electric system or electronic (micro-chip) blasting systems.

63.9 Protection against fly rock

63.9.1People should be removed from the blasting area to a safe place where they will be protected from projectiles or flying rock.

63.10 Testing the firing circuit

63.10.1Testing the electric firing circuit for continuity is necessary in order to prevent misfires. Testing consists of measuring the circuit resistance with an ohmmeter of an approved type.

63.10.2Testing should be done from the firing point and if any fault is detected it should be rectified before the blast.

63.11 Precautions after Blasting

63.11.1Adequate ventilation should be provided and atmospheric monitoring carried out to ensure the removal of all hazardous gases before re-entry.

63.11.2The shot firer and assistant should check the face and make safe before allowing work to proceed.

63.11.3Before mucking operations the responsible person shall check to ensure that safety precautions have been carried out.

63.11.3Vehicles used for the transport of explosives or detonators should be regularly inspected and maintained to check that they do not present any risk to the materials carried.

64. Monitoring of the tunnel atmosphere

64.1 General

64.1.1A summary of the most commonly encountered atmospheric contaminants and their properties is given below. This should be used as a guide when controlling the risks from these hazards.

64.1.3Monitoring for oxygen concentration should be carried out routinely by the use of electronic atmospheric monitoring equipment. Where the presence of gaseous contaminants is foreseeable, continuous monitoring by electronic equipment should likewise be carried out.

64.1.4At range of gases including methane, hydrogen sulfide and radon dissolved in ground water may enter the tunnel.

64.1.5Ventilation should be used to dilute contaminants and to prevent the build up of an irrespirable atmosphere.

64.2 Summary of The Most Commonly Encountered Atmospheric Contaminants

Gas	Hazard	Principal sources

Carbon monoxide	Toxic and potentially explosive	Internal combustion engines and explosives
Carbon dioxide	Asphyxiant	Occurs naturally, explosives, welding and internal combustion engines.
Nitrogen monoxide Nitrogen dioxide	Toxic Toxic	Internal combustion engines, explosives and welding
Ammonia	Toxic and potentially explosive	Explosives
Hydrogen sulphide	Toxic and potentially explosive	Occurs naturally and may occur where sewage is present
Sulphur dioxide	Toxic	Occurs naturally
Methane	Potentially explosive (and asphyxiant)	Groundwater, decay of organic matter, urban domestic gas supplies, Carboniferous rocks
Propane Butane Acetylene	Potentially explosive (and asphyxiant)	Leakage from compressed gas cylinders Do
Oxygen deficiency	Asphyxiant	Natural/induced
Oxygen enrichment	Increased fire risk	Leaks from cylinders
Petrol/diesel vapour	Potentially explosive	Spillage
Ozone	Toxic	Welding
Radon	Radioactive	Ground

65. Rolling stock

65.1.1 Muck skips are usually designed for either tipping or hoisting. When skips are hoisted they should be regarded as a piece of lifting equipment and designed and used accordingly. Periodic examination and testing shall take place as for lifting tackle. When skips are tipped, provision must be made to lock them during travelling.

65.1.2Purpose-built skips and materials bogies must incorporate restraint of the intended load to prevent, for example, a box or stack of segments from sliding off and fouling walkways. The design of the vehicle should be inherently stable when loaded, even on poor track and with rough handling.

65.1.3Certain high-capacity cars have flight bar conveyor mechanisms to discharge the muck at one end at a controlled rate. Where these types of cars are used, special attention shall be given to the manufacturer's recommendations concerning track laying, layout of curves and loading across overlap points to minimize the potential for derailment.

65.1.4Personnel can be transported if purpose-built man-riding cars are used. These shall be attached as near the locomotive as possible with a safety chain in addition to the coupling. A man-riding car must be constructed with proper seats, side bars and a crash cage. It shall be constructed to withstand derailment with minimum injury to passengers. It is recommended that the largest practical wheel size be used with a suspension system on man-riding cars.

65.2 Transportation of material

65.2.1Cars carrying pipe and rail shall be properly loaded for safe passage through the tunnel. The load shall be kept within the side limits for the car. Loads projecting over the sides are dangerous to men working in the tunnel. If wide' loads are transported, a special care shall be ensured in the operation of the train with ample warning to the workmen along the track to ensure a safe journey.

65.3 Transport of Employees

65.3.1No one shall be allowed to ride on front steps of a loco or on a coupling. None other than trainmen shall ride the dump cars in going to and from work at change of shift or at any other time. A safe and smooth walkway system shall be provided for pedestrians

65.4 Roles and Competencies

65.5 The Locomotive Driver shall:

65.5.1Be responsible for ensuring that all loads are properly secured before a loco movement is commenced

65.5.2Be responsible for ensuring that identified defects are entered in the Daily Loco Checklist.

65.5.3Have received a minimum of 04 hours familiarisation training in accordance with local arrangements and have been assessed as competent.

65.5.4Successfully complete third party approved training.

65.5.5Attend local Tunnel Induction training.

65.5.6Be appointed as a Locomotive Driver in writing by site management; records of this are to be held on site in accordance with BS6164:2011

65.5.7Be fully briefed and conversant with all relevant local procedures.

65.5.8Be aware of track walking procedures, persons walking on the track, persons working on the track, track possessions etc.

65.5.9Check points are fully and correctly engaged before entering any crossings.

65.5.10Ensure that when propelling (breasting) the train into the tunnel, this is carried out in accordance with BS 6164: 2011– the driver must have sufficient visibility of the track ahead to be able to stop within his field of view.

65.5.11Be trained in the use of derailers, wheel blockers, Manchester gates etc.

65.5.12Be trained in the use of communications systems and all relevant procedures and protocols.

65.5.13Be trained in the use of fire extinguishers and relevant firefighting equipment.

65.5.14The training will be supported by refresher training and any other briefings as deemed appropriate by Employer / Engineer.

65.6 Operations Board Controller (OBC)

65.6.1The Operations Board Controller shall:

65.6.2Have been assessed as competent to carry out all activities associated with the role.

65.6.3Be responsible for maintaining a record of all hazardous materials contained in each load so that appropriate information is available should an emergency arise.

65.6.4Have experience of a similar role.

65.6.5Successfully complete Third party training.

65.6.6Attend local Tunnel Induction training and be trained in the requirements of this Best Practice Guide.

65.6.7Be fully briefed and conversant with all relevant local procedures.

65.6.8The contractor will ensure that there is adequate cover for this post at all times.

65.6.9The contractor will ensure that there are local instructions in place which cover key issues including:

- Use of the Operations Board
- What information is recorded
- How and where information is recorded
- Priority assignment on California crossings

65.6.10 Note: Where there is a permanent single line running arrangement (i.e. only one train on the track at any given time), it will not be necessary to engage an OBC, but ONLY if the contractor has put in place adequate arrangements to protect the train and those working in the tunnel.

65.7 Flagman

65.7.1Where a flagman is engaged (e.g. at a California crossing) they shall:

65.7.2Receive a minimum of 4 hours familiarisation training.

65.7.3Have been assessed as competent to carry out all activities associated with the role.

65.7.4Successfully complete Third party training.

65.7.5Attend local Tunnel Induction training.

65.7.6Be fully briefed and conversant with all relevant local procedures.

65.7.7Be trained in the application of this Best Practice Guide. This training will be supported by refresher training and any other briefings as deemed appropriate by local management.

65.8 Back-up operator / slinger

65.8.1The Back-Up Operator shall:

65.8.2Have experience of working within and on a TBM.

65.8.3Have receive a minimum of 4 hours familiarisation training.

65.8.4Have been assessed as competent to carry out all activities associated with the role.

65.8.5Have Successfully Complete Third party training.

65.8.6Attend local Tunnel Induction training.

65.8.7Be fully briefed and conversant with all relevant local procedures.

65.8.8Carry out all duties as defined in local working instructions.

65.8.9Operate the Manchester gate as and when required.

65.8.10Supervise the ingress / egress of personnel movements into and from the stationary man rider.

65.9 Pre-use Checks

65.10 Locomotives and rolling stock

65.10.1Locomotives and rolling stock scheduled for use will be inspected by the driver at the start of each day to ensure that it is fit for service and capable of delivering safe operation. This inspection will include a brake functionality test (as required by BS 6164:2011).

65.10.2The Daily Loco Checklist (DLC) remains on the locomotive; any defects or issues identified by the driver are to be logged at the first available opportunity prior to use.

65.10.3Unserviceable safety critical equipment must be removed from use as soon as a defect is identified.

65.10.4The DLC is to be checked daily by the locomotive fitters who will:

i) Review entries

ii) Carry out and record any necessary repairs

iii) Schedule or carry out routine maintenance as appropriate.

iv) Report recorded defects to the OBC.

v) Complete a record of all work carried out on the locomotive.

vi) The driver will test on board communications and confirm both the locomotive and communications are operational with the OBC prior to receiving instructions at the start of each shift.

65.11 Train-borne plant

65.11.1The following items are to be checked by fitters at predetermined periods:

i) The engine compartment fire suppression system.

ii) The hydrostatic transmission and retarder.

iii) Pneumatic wheel brakes.

Iv) In addition, couplers and chains (for the rolling stock) are to be checked by drivers before each movement.

v) Should any of the above items be found to be defective during checking, the loco fitter should be consulted and appropriate remedial action taken.

65.12 Training and briefing

65.12.1The following are to be applied:

i) Locomotive drivers must have received familiarisation training on the locomotive they are driving and have been assessed as competent to drive.

ii) All personnel on site are to be briefed in the communication procedure so that they are aware of actions to take in the event of an emergency.

65.13 Locomotive driver aids

65.13.1The following checks are to be carried out by the loco driver at the beginning of each shift:

i)CCTV system is functionally tested and camera lens wiped clean.

ii)Functional brake test.

iii)Test that lights are operational and showing correct direction of travel

iv)Communication system is functionally tested

v)Locomotive Electronic Control is tested

vi)Horns tested

vii)Note: The locomotive should not be used if any of these systems are defective.

viii)Rail Movement Control Centres (RMCCs)

ix)Where RMCCs are in use, the following will apply:

x)Where the RMCC is equipped with CCTV cameras they are to be tested and confirmed to be working correctly by the OBC at the beginning of each shift

xi)The communication system in the RMCC is tested and confirmed to be working correctly by the OBC at the beginning of each shift

65.14 Communications and signalling equipment

65.14.1The following will be applied for communications and signalling equipment:

i)The tunnel radio system is to be functionally tested by the OBC at the start of each shift

ii)The CCTV system (where installed) is to be functionally tested by the OBC at the start of each shift

iii)Signs are to be installed establishing the maximum speed (e.g. California crossings)

iv) “Trains have priority” signs are to be installed at all level crossings

v)Cameras are to be installed, tested, certified and working at both ends of all crossings

vi)The following horn blasts are to be used in the tunnels to designate the imminent action of a loco:

3 blasts	Moving outbye
2 blasts	Moving inbye
1 long blast	Stopped

65.14.2A continuous blast is to be given when the driver identifies personnel ahead in the tunnel who have not acknowledged and moved out of the way of the train.

65.15 Vehicle Movement Safety

65.16 General

65.16.1The OBC will be located within a clearly defined area and will be in overall charge of all train movements between the yard / loco stabling area and TBM

65.16.2The Back-Up operator will be in control of all train movements within the TBM for segment movements, train unloading movements and personnel movements

65.16.3Where flagmen are used, they will direct all trains as required and always under the control of the OBC and yard staff to ensure safety

65.16.4In addition to being appropriately trained and certificated, locomotive drivers will have read the operations manual for the locomotive they are operating.

65.16.5Passengers are not permitted in the locomotive cab at any time EXCEPT When the driver is under training and the passenger is the instructor

65.16.6Apart from walking along approved safe walking routes, personnel transport into the tunnel will be by man rider only. Travelling on any other part of the train is prohibited

65.16.7Carrying work equipment or materials in the man rider is prohibited EXCEPT for surveying equipment

65.16.8Pedestrians are not allowed into tunnels unless carrying out surveys, track inspections, maintenance etc. In such cases, access is only permitted under the control of the OBC and with an appropriate Permit to Work in place

65.16.9Pedestrians are only permitted to cross the track on marked safe walking routes

65.16.10All tools and materials are to be kept clear of the track at all times UNLESS work is being carried out on the track. All such work will be carried out under (and in accordance with the requirements of) a Permit to Work issued by the OBC

65.16.11The tunnel will be equipped with a communication system allowing radio communication between the loco operator and other tunnel personnel

65.16.12Where local noise restrictions exist, alternative arrangements will be put in place for noise restricted times

65.16.13The loco driver will comply with the traffic lights which are controlled by the appointed Back-Up Operator

65.16.14Parked trains or trailers will be individually air braked and wheel scotches placed under wheels of uncoupled rolling stock. Scotches may be of wooden or metal construction. Dog chains (or similar) are not permitted

65.16.15Brakes and scotches / rail clamps are to be applied on static uncoupled trailers to prevent them running away

65.16.16The loco driver will be responsible for carrying out checks of the loco to ensure that it is fit for service. Checks are to be carried out in accordance with the mechanical and operational manual checklists provided by the loco manufacturer

65.16.17The loco driver will be responsible for carrying out checks to ensure that there is adequate clearance along the route for the train to travel safely without risk of colliding with obstructions; adherence to gauge limits must be ensured

65.16.18Tampering with couplings is forbidden

65.16.19Prior to inbye movements, both the loco driver and the yardman will check train loads to ensure they are secure

65.16.20Engines should be switched o• when not required to reduce unnecessary fuel consumption and emissions

65.17 Tunnel movements

65.17.1The following are to be applied for all tunnel movements:

65.17.2The maximum train speed within the tunnels is to be displayed on signposts at appropriate locations; this speed limit is an absolute and is not to be exceeded at any time

65.17.3The maximum speed for traversing a California crossing will be 5km/h; warning notices are to be placed 60m either side of the crossing instructing loco drivers to slow down

65.17.4Movement direction lamps at the front and rear of the train are to be illuminated at all times to indicate the direction of travel. The lights should show white at the front of the train in the direction of travel and red at the rear when a travel direction is selected

65.17.5When passing personnel in the tunnel the train driver will sound the horn to advise of his approach; the personnel concerned are to wave a hand to acknowledge that they are aware of the approaching train

65.17.6Walking in the tunnels is not encouraged. However, it is acknowledged that teams are required to work in the tunnels at certain times. When doing so they will protect their working area with advanced warning lights which may be obtained from the OBC; these warning lights are to be placed 60m either side of the work area. Such works may only be carried out under a controlled Permit to Work (PTW) in accordance with local arrangements

65.17.7PTWs will be held by the OBC who is responsible for briefing all relevant personnel on the content of the PTW including the nature of the work and number of personnel in the work group

65.17.8Loco drivers are to be continuously advised of all PTWs that are introduced / signed of during a shift

65.17.9PTWs are only signed on when the work is complete, the site has been made safe and all members of the working group are accounted for. A PTW may be passed to a subsequent working group, but the names of all outgoing and incoming personnel must be recorded in accordance with the local PTW procedure

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65.17.10There will be intermediate stop points (crossings and booster locations) where workers will be able to alight, but always within the confines of the PTW and under the control of the OBC

65.17.11All derailments must be reported for appropriate investigation, identification of trends and implementation of corrective actions. Should a derailment occur, local procedures for re-railing are to be followed

65.17.12 No work is permitted to start in the tunnels until a Permit to Work is in place.

65.18 Uncoupling trailers

65.18.1The following actions are to be taken when uncoupling wagons and trailers:

i) Apply parking brakes and put scotches / chocks in place before uncoupling; no unbraked vehicles should be allowed in the tunnels

ii) Install the scotch / chock on the downhill side of the trailer; if ‘downhill’ is not clear (or if the vehicle is on level ground), clamps should be applied at both back and front

iii) Verify that brakes are still on

iv) Disconnect lighting, CCTV and telephone cables

v) Disconnect hydraulic brake lines ensuring that all supplies are isolated as per the operational instructions for the vehicle

vi) Check again that the brake is applied and that scotches / chocks are firmly in place

vii) Disconnect the safety chain and then uncouple the vehicle

viii) The vehicle may now be left

65.19 Coupling trailers

65.19.1The following actions are to be taken when coupling / uncoupling wagons and trailers:

i)Where Manchester gates have been installed, they must be kept closed until the fully coupled train is ready to move into the tunnel

ii)Ensure scotches / chocks are in place before commencing coupling

iii)Connect vehicles using couplers and secure the safety chain. Safety chains must be loose enough to allow the trailers to travel round bends in the tunnel but not so long that they are able to touch the track

iv)Connect the hydraulic brake lines. Care must be taken to ensure that connections are clean and seated properly with no leaks. Visual / audible checks are to be carried out. Hydraulic pipes must not touch the track

v)Connect lighting, CCTV and telephone cables

vi)Once all the above connections have been completed the scotches/chocks may be removed and the vehicle can then be moved

NOTE: Vehicles should not be moved until the driver has confirmed that the train is fully functional

65.20 Trains entering the TBM backup

65.21The following rules are to be applied for all trains entering the TBM backup:

i) The train must signal its arrival by sounding the horn in accordance with local procedures and stop in front of the Manchester gate

ii) No train movements are to take place at the rear of the TBM until the Back-Up operator on the TBM has confirmed to the loco driver that:

iii) All personnel are clear of the track

iv) No personnel are attempting to board or disembark from the man rider

vi) When the Manchester gate is open the free access will be indicated with a green light indicating permission to move in bye. The Manchester gate opens only through hand operation controlled by the appointed Back-Up Operator

vii) The locations for stopping the train will be marked on the gantries and indicated by a switch of the traffic light to red. Entry to the rear of the TBM must be at creeping speed

viii) One long blast on the loco horn will be used to indicate that the loco has stopped moving and brakes have been applied. Personnel entering or leaving the man rider will only do so when given the all clear by the appointed Back-Up Operator

ix) Once the train is unloaded it will leave out bye the TBM backup at creeping speed once a blue light is displayed

x) The Appointed Back-Up Operator will check before permitting movement of the man rider that no personnel are attempting to board the train before it departs

xi) Once the operation is complete, the loco will not be permitted to move of until the appointed Back- Up Operator has indicated to the loco driver that no personnel are attempting to board or disembark from the man rider

xii) Immediately prior to moving of the loco driver will sound three blasts of the loco horn to indicate that the loco is moving out bye

65.22 Track inspections and maintenance

65.22.1Track inspection and maintenance activities will be completed by a dedicated track maintenance team on a scheduled basis (additional works will be carried out as necessary). This work will be compliant with the following

65.22.2Local processes will be in place to detail routine maintenance, planned preventive maintenance, breakdown repair protocols and re-railing procedures

65.22.3Local processes should include information on when train movements are to be stopped for emergency maintenance (e.g. submerged track, build up of grease in points etc.)

65.22.4The Designated Person / Persons undertaking track maintenance must be known / communicated to drivers and other tunnel personnel

65.22.5All track maintenance and inspection activities must be carried out under a Permit to Work issued by the OBC

65.22.6Any track defects are to be reported to the OBC as soon as they are identified, by whoever identifies them

65.22.7In order to avoid derailments, the track will be inspected visually during the routine maintenance shift. Bolts will be retightened as necessary

65.22.8In order to avoid derailments, the track will be inspected visually during the routine maintenance shift. Bolts will be retightened as necessary

65.22.9In the event of a derailment, any affected track must be fully inspected and appropriate remedial action taken where necessary

65.22.10Spillage of materials that may affect safe train operation will be reported to the OBC who will make arrangements with the Tunnel Superintendents for the spillage to be cleared as soon as practical

65.22.11Track geometry will be checked (gauge and rail inclination). Special attention is to be paid to track at the portals and sections with very small radius (<40-50m) in order to avoid derailments

65.22.12Checks are to be carried out on the different geometrical parameters of the California crossings in order to avoid derailments

65.22.13Following any change to the track, gauge is to be checked and movements only permitted when confirmation is given that gauge is correct

65.22.14Personnel working on the railway will wear the appropriate PPE and will operate under a Permit to Work in accordance with the local procedure

65.22.15The invert is to be kept clear of hazards as far as reasonably practicable; the underside of the rails must be visible above any muck in the invert

65.23 Loading and unloading (TBM)

65.23.1The loco driver must make sure that a competent person supervises the loading/unloading of the loco. This includes ensuring that the load is:

i) Correctly loaded

ii) Secured to prevent movement

iii) Within the maximum load limit for the vehicle

iv) Within the maximum gauge limit for the track

v) The loco driver must sign the loading note and train log book to certify the load as being correct and safe to travel.

65.24 Loco and train maintenance

65.24.1The locomotives and trailers are to be checked and maintained according to the manufacturer’s operation manual

65.24.2Maintenance is only to be carried out by competent personnel appointed in accordance with the requirements of BS 6164: 2011

65.24.3A record is to be maintained of all maintenance and repairs carried out on the loco and trailers

65.24.4Where a defect has been identified that makes the loco / train unfit for safe use, it is not to be used until appropriate remedial action has been taken

65.24.5Maintenance and repairs are only to be carried out in designated areas. It is forbidden to carry out any maintenance / repair activities outside the designated areas EXCEPT when the train has failed in the tunnel and must be repaired before it can be moved

65.24.6The design of cab or driver-containment enclosure should allow reasonable freedom of movement for the driver.

65.24.7CCTV can be used to allow the driver a good view ahead.

65.24.8 Visibility should be adequate for safe driving of the train.

65.24.9The lights should automatically show white in the direction of travel and red to the rear of the locomotive when a travel direction is selected.

65.24.10A daily functional brake check should be carried out by the regular drivers

65.25 Use of new plant

65.25.1The Tunnel Manager must approve any new plant that is to be used in the tunnel

65.25.2All plant must be recorded on the asset register and regularly checked and maintained in accordance with manufacturers recommendations

65.25.3Maintenance records are to be kept for the new piece of plant, along with records for all other plant currently in use

65.26 Refuelling

65.26.1Refuelling will be at a dedicated and segregated area as far from the works area as practicable

65.26.2The refuelling point will be in open air; fuel will be obtained using a hose and trigger nozzle from a tank located on the surface

65.26.3The trigger is to be locked on when not in use

65.26.4Bulk spill containment, fire extinguisher and oil binder will be in place as close to the fuelling point as practicable

65.26.5Leaving the nozzle unattended while refuelling is underway is prohibited in all circumstances; if it is necessary to leave the refuelling point the nozzle must be removed and replaced in its' retainer on the pump.

65.27 Communications protocols

65.27.1OBCs are to have overall responsibility for management of communications. In all cases:

65.27.2Instructions issued by the OBC are to be followed UNLESS it is believed that the instruction may present a risk

65.27.3Instructions issued by the OBC take precedence over all other instructions relating to the railway UNLESS it is believed that the instruction may present a risk

65.27.4All communications relating to incidents (including near misses) are to be recorded by the OBC

65.27.5When communicating alphanumeric, the phonetic alphabet is to be used at all times.

65.28 Emergencies

65.28.1Emergency response procedures are to be followed.

65.29 Tunnel vehicle maintenance

65.30 General

65.30.1All rail vehicles are required to be maintained in accordance with a maintenance plan.

65.30.2The organisation responsible for management of the construction railway must obtain the maintenance documentation which details vehicle maintenance activities from the vehicle manufacturer. Maintenance activities include all activities necessary such as inspections, monitoring, tests, measurements, repairs, replacements and adjustments.

65.30.3Maintenance activities are split into:

65.30.4Preventive maintenance; scheduled and controlled

65.30.5Corrective maintenance (repair)

65.30.6Maintenance documentation should include the following:

i)Component hierarchy and functional description. The hierarchy should list all the items belonging to the vehicle and use an appropriate number of discrete levels; the lowest level shall be a replaceable unit

ii)Schematic circuit diagrams, connection diagrams, wiring diagrams, pneumatic schematics and lubrication diagrams

65.30.7The maintenance plan.

65.30.8This is the structured set of tasks that include the activities, procedures, means and the working time required to carry out the maintenance task

65.30.9It contains a description of the maintenance activities which include the following:

i)Disassembly / assembly drawings & instructions necessary for correct assembly / disassembly of replaceable parts

ii)Maintenance criteria

iii)Checks and tests

iv)Tools and materials required to undertake the task

v)Consumables required to undertake the task

vi)Personal protective safety provision and equipment

vii)Necessary tests and procedures to be undertaken after each maintenance operation before re-entry in to service of rolling stock

viii)Checks on lifting / jacking point

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ix)Troubleshooting (fault diagnosis) manuals or facilities for all reasonably foreseeable situations. This includes functional and schematic diagrams of the systems or IT-based fault-finding systems

65.31 Maintenance plans – General

65.32 Relevant to all vehicles

65.32.1Vehicle maintenance activities are only to be carried out by appropriately qualified personnel.

65.32.2The maintenance plan shall detail all maintenance activities to be undertaken on the applicable type of rail vehicle

65.32.3The maintenance plan should include all maintenance activities that must be carried out to ensure that the rail vehicle(s) continue to conform to the safe limits defined by the vehicle manufacturer

65.32.4The maintenance plan shall, as a minimum, include the following:

i)Maintenance requirements including the safety conditions to ensure the rail vehicle can be safely worked on for each task

ii)A maintenance schedule which defines the periodicity at which each item shall be actioned

iii)An inspection programme for regular inspection of the vehicle to confirm that it is safe to continue in service

iv)Definitions of the appropriate actions to be taken to ensure that all systems and equipment continue to operate safely over the full range of environmental conditions, particularly in snow, flood, freezing or abnormal heat

v)Technical instructions that define actions required to ensure that a vehicle can be hauled safely when inoperative (e.g. the adjustment, isolation, removal or addition of some components or the imposition of a restrictive maximum speed to travel

vi)The minimum engineering maintenance facilities that are necessary for the maintenance to be carried out (e.g. pits, major equipment, covered work areas)

vii)The minimum level of competencies required for the staff carrying out the specified maintenance tasks

viii)Each rail vehicle shall be maintained so that the prescribed tolerances for all components and assemblies are not exceeded throughout the life of the vehicle

65.33 Maintenance Plans – Specific

65.34 Relevant to all vehicles

65.34.1The maintenance plan shall, as a minimum, set out the requirements for the following items of the wheelset and constituent parts:

i) Relative movement of wheels, axles, tyres and axle mounted equipment

ii) Cracks and fractures

iii) Dimensions affecting running safety:

iv) Minimum wheel diameter

v) Tolerance between diameters of wheels on the same axle

vi) Back to back dimensions

vii) Flange and tread profile

viii) Wheel tread surface damage

ix) Wheel flat limits

65.35Brakes

65.35.1The maintenance plan should detail that the components of rail vehicle brake systems are maintained at appropriate periods

65.35.2Maintenance activities and periods must be designed to ensure that the brake systems function correctly and safely between maintenance activities

65.35.3Functional tests of the brake systems should be carried out to ensure that they are operating correctly. This should take into account the environment in which the vehicle is being used

65.36.4The functional brake tests should show that the systems respond to graduated brake application demands, up to and including an emergency stop

65.35.5Brake cylinder pressures should be checked to confirm that pressure increases in proportion to the controller movement

65.35.6Appropriate functional brake tests should be undertaken whenever components of the brake system are replaced and reconnected or following component repair, renewal or disconnection

65.35.7If a fault in a brake system or component is revealed by either the maintenance or functional brake tests, details of appropriate remedial action should be available in the maintenance plan

65.35.8The maintenance plan shall, as a minimum, set out the requirements for (where fitted):

65.35.9Brake disc integrity, condition and dimensions

65.35.10Brake pad and brake block integrity and dimensions

65.35.11Brake rigging (pins, bushes and associated equipment)

65.35.12Integrity of operating devices, reservoirs, hoses, cocks, pipework, safety loops and associated equipment

65.36 Rail vehicle structure

65.36.1Vehicles shall be maintained so that the body and running gear remain structurally sound and safe

65.36.2Vehicles shall be maintained so that the connections between the body and running gear remain structurally sound and safe

65.36.3Prescribed tolerances must be checked and maintained – this includes all components, assemblies and systems

65.37 Speed indicating equipment

65.37.1The maintenance plan must include a method for testing speed indicating equipment – this must include acceptable tolerances. If readings are outside this tolerance, the speed indicating equipment must be corrected before the vehicle re-enters service

65.37.2The speedometer test shall cover the whole range of the speed indicating equipment. The test, shall cover:

i) The accuracy of readings

ii) Intermittent or jerky operations

iii) Clarity and cleanliness of indicators and correct operation of associated lighting

iv) The integrity of connections and components that cannot be included in the equipment test

v) As a minimum, speedometer and speed control system testing shall be undertaken in the following circumstances:

vi) When a wheelset has been replaced

vii) When a wheelset has been re-profiled

viii) When the speed indicating equipment, speed control system or their components have been disturbed, adjusted, repaired or replaced

ix) When there has been a report of a malfunction of the speed indicating equipment or speed control system

x) When there has been a report questioning the accuracy of the speed indicating equipment or speed control system

xi) Following an allegation of exceeding the speed limit

65.38 Train-borne signalling and communications equipment

65.38.1All signalling and communications equipment shall be managed to ensure that its integrity and performance remains compliant with its specification

65.38.2The maintenance requirements for each item of train-borne signalling and communication equipment shall be developed to control the risks which would arise from failure of any part the system

65.38.3The maintenance requirements for train-borne signalling and communication equipment shall be documented

65.38.4Maintenance requirements shall form part of the rail vehicle maintenance plan for the vehicle

65.38.5The procedures to ensure configuration control of any software and hardware shall also be documented in the maintenance plan

65.39 Headlights

65.39.1The maintenance plan shall set out a method for testing the functionality and alignment of the headlights

65.39.2 Testing of the headlights shall be carried out at a periodicity to reflect the use and working environment of the vehicle

65.39.3 Other equipment

65.39.4 Consideration shall be given to the following during the production of the maintenance plan. The list is not exhaustive or necessarily representative of all types of vehicles used on construction railways.

- Buffers:
- Heights
- Integrity and condition
- Greasing
- Draw gear:
- Dimensions
- Rubber condition
- Integrity and condition
- Operation
- Primary and secondary suspension:
- Spring integrity, rules for changing
- Linkage wear, dimensional limits
- Suspension settlement
- Damper integrity
- Suspension tube bearings:
- Condition, integrity and security of components and installation
- Bearing float

65.39.5 Traction and auxiliary generators, alternators, other electrical machines and electrical equipment cases:

- Integrity and security
- Earthing condition and integrity
- Presence, condition and cleaning of all safety labelling
- Cleaning:
- Ventilation ducts
- Bogies and underframe equipment
- Front end and windscreens
- Final drives / transmissions:
- Security to body or bogie frame
- Condition checks of safety critical items
- Lubrication
- Safety loops
- Presence and security of balance weights, where fitted

- Internal combustion engines:
- Security and condition of mountings
- Checks for leakage of fluids
- Integrity of shaft couplings
- Engine safety systems (for example, overspeed, crankcase explosion)
- Emissions (compliant with specification)
- Power systems (including associated protection systems):
- Integrity and security
- Earthing condition and integrity
- Presence, condition and cleaning of all safety labelling
- Hydraulic and pneumatic systems:
- Condition and integrity of hoses, pipework, valves, etc
- Fire protection systems:
- Integrity and condition
- Currency of certification
- Lighting Systems
- Emergency Facilities:
- Emergency lighting capacity
- Other emergency and safety equipment provided

65.40 Tunnel Track maintenance

65.41 Roles and Competencies

65.42 General

65.42.1This Section defines the general duties and competence requirements of personnel with a role in maintaining track, crossovers and associated equipment on the construction railway. It should be referred to when clarification is required on specific responsibilities.

65.42.2The role titles used below may not necessarily correspond with job titles in the Contractors' organisation, however the Contractor will be expected to demonstrate that the responsibilities listed below have been discharged by a competent person with sufficient knowledge and experience.

65.43Mechanical Superintendent

65.43.1 The Mechanical Superintendent (or equivalent) shall:

- Be responsible for ensuring that a track maintenance programme is in place covering all tunnel railways track, California crossings, sleepers, fishplates and associated components
- Ensure that all track and associated equipment is inspected on a regular basis and to a pre-determined frequency.

- Ensure that a forward-looking programme is in place covering the inspection and maintenance of track and associated components
- Successfully complete Third-party training.
- Attend Tunnel Induction training and the Contractor's Track Maintenance Procedures
-

65.44Track Inspector

65.44.1The Track Inspector shall:

- Be responsible for carrying out inspection and maintenance of track, California crossings, sleepers, fishplates and associated components to the standards prescribed in the contractors' track maintenance procedures
- Have received training and hold the necessary competence in the inspection and maintenance of track
- Be responsible for maintaining a record of all track inspections
- Successfully complete third-party training
- Attend Tunnel Induction training and the Contractor's Track Maintenance Procedures

Note: Tracks laid in the TBM are to be inspected immediately after installation. This can be carried out by an inspector or by any suitably qualified member of the track gang.

65.45Operations Board Controller (OBC)

65.45.1The Operations Board Controller (OBC) shall:

- Provide protection for track inspectors whilst undertaking track maintenance and inspection activities
- Communicate with all locomotive drivers when track inspectors are working in the tunnel
- Arrange the safe system of work so the location of track inspectors working in the tunnel is known

Note: The Contractor shall ensure that there is continuous cover for this post at all times. Where there is a permanent single line running arrangement (i.e. only one train on the track at a given time), it will not be necessary to engage an OBC, but only if the contractor has put in place adequate arrangements to protect the train and those working in the tunnel through a permit to work system.

65.46 Tunnel Tracks Work Permit

65.47Arranging the Track Works Permit

65.47.1Before any track inspection or maintenance work is carried out, the Track Inspector (or other person undertaking work on or about the track) shall report to the OBC to arrange for a Permit to Work – the Tunnel Track Works Permit - to be issued. The OBC will raise the Tunnel Track Works Permit which will specify, as a minimum:

- The location of the works to be carried out

- The names and details of persons undertaking the track inspection and maintenance works
- The means of communication between the Track Inspector and the OBC during the works
- The duration of the work to be carried out
- The nature of the work and any requirements for train movements to be suspended or restricted (for example a temporary speed restriction during or after the works)
- The time at which the permit will be closed

65.48 Issuing the Tunnel Track Works Permit

65.48.1The OBC will sign the Tunnel Track Works Permit and the permit will then be approved by the Mechanical Superintendent, Tunnel Superintendent or other nominated person named in the contractor's procedure.

65.48.2A permit to work must be issued when entry to the tunnel is required to carry out track inspection or maintenance duties, irrespective of whether train movements are taking place or not.

65.49 Cancelling the Tunnel Track Works Permit

65.49.1Upon completion of the works, the Track Inspector will notify the OBC that:

a) All personnel are clear of the track and are in a position of safety

b) All tools and equipment have been removed from the track

c) The track is safe for normal operations to be resumed, unless operating restrictions (for example a speed limit) need to remain in place

65.50 Tunnel Track Maintenance Works

65.51 Track Inspection Programme

65.51.1The Mechanical Superintendent will develop a forward-looking programme of track inspections covering a 12 month period

65.51.2The Tunnel Track Inspection Programme will detail the sections of track to be inspected on a particular track patrol, identified by segment ring number

65.51.3The Tunnel Track Inspection Programme will also define any components, for example California Crossings, which require specialised inspections other than a normal visual check

65.51.4Additional visual track inspections shall be undertaken following any reported rough ride, derailment or track defect observed by the loco drivers

65.51.5The Track Inspection Programme shall ensure that every section of track is inspected at least once every week.

65.51.6The track inspection will be undertaken on foot under adequate lighting conditions. Where 2 or more lines exist on a stretch of track, each line shall be walked separately

65.51.7The track inspection will include a visual examination of the condition of the track and identification of any rail defects, pitting, track spread or rail buckles

65.51.8At regular intervals defined in the contractor's track maintenance procedure, the track gauge will be measured and recorded using a calibrated track gauge. Where the gauge is outside the prescribed tolerance of 900mm +/- 25mm, the OBC shall be notified immediately

65.51.9Any significant defects in the track, for example broken rail, track spread, buckled rail shall be reported to the OBC and train movements suspended until repairs have taken place or the Mechanical Superintendent has confirmed that it is safe for passage of trains

65.51.10Track inspections shall be recorded on a proforma defined in the contractor's track maintenance procedure

65.52 Track Bed

65.52.1The track inspection shall include a visual examination of the sleepers & track bed, which will take place as part of the track inspection, including:

a) Loose or missing Pandrol clips, or other rail to sleeper fixture
b) Broken or damaged sleepers
c) Loose or damaged "anchors" which connect the sleepers to the concrete segments
d) Build-up of oil, waste material, or excessive water on or around the track bed

65.53 Fishplates and bolts

65.53.1The track inspection shall include a visual examination of the fishplates and fishplate bolts, which will take place as part of the track inspection, for:

a) Loose or missing fishplate bolts
b) Broken or cracked fishplates
c) Condition of the grease between the fishplate, the fishplate bolts and rails

65.53.2Where "dry joints" are detected, the fishplate bolts will be loosened, grease will be applied to the rail edges of the fishplate and bolts, then the bolts shall be re-tightened using a torque spanner to the setting prescribed in the contractor's track maintenance procedure.

65.54 California Crossings

65.54.1California Crossings represent a higher risk of derailment than plain line track and therefore an enhanced inspection regime shall be adopted.

65.54.2The frequency of inspections of California Crossings and associated switches shall be at least twice weekly. Inspections shall be undertaken by the Track Inspector and recorded on the contractor's record of track inspection

65.54.3Inspections of California Crossings shall additionally be undertaken following any report of rough ride, derailment or other malfunction reported to the OBC.

65.54.4Inspections of the California Crossing shall include:

65.54.5A visual check of the crossing, including the switch blades, for signs of damage, corrosion, wear and obstructions in the switch blades

65.54.6A check of the movement of the switch blades from normal to reverse positions and back

65.54.7Application of mineral or bearing grease to the moving parts of the switch blades

65.54.8A visual check of the mechanical parts of the California Crossing, to include the tie rods, stabilisers, section connections, point levers and spring mechanisms

65.54.9In the case of a post-derailment inspection, the Track Inspector shall observe train movements in both directions over the crossing to check the operation

65.55 Records of Track Inspections

65.55.1A bespoke record of track inspections and maintenance shall be kept for every track inspection. The completion of the record of track inspection shall be defined in the contractor's track inspection procedures

a) The record of track inspection shall include the following information, as a minimum:
b) The date and time of the track inspection
c) The name and signature of the Track Inspector
d) The section of track patrolled – identified by segment ring numbers
e) The line patrolled, where more than one exists
f) Nature of any defects identified
g) Any work that was identified from the inspection that could not be completed at the time
h) Overall remarks on the condition of the track and any potential defects which could arise before the next scheduled inspection

66. Pathways

66.1 Tunnels the walk ways shall be placed to the side of track as per the BS 6164:2011

66.2 Tunnels, shelter places for workmen shall be provided at suitable intervals during hauling operations.

66.3 Elevated walkways from which a fall is possible should be provided with suitable guard rails and toe boards

66.4 A walkway should be provided within a minimum clear space of 2000 mm high and 900 mm wide, with a walking surface not less than 430 mm wide.

66.5 Water should not be allowed to stand or accumulate at or above walkway level at any point

67. Visitors to site

67.1 No visitor is allowed to enter the site without the permission of the Employer / Engineer . All authorised visitors must report to the site office. The Contractor shall provide a visitor's helmet (White helmet with visitor sticker) and other PPE such as Safety Shoes, reflective jacket, respiratory protection etc. as per requirement of the site.

67.1.1 All Visitors shall be accompanied at all times by a responsible member of the site personnel.

67.1.2 The contractor shall be fully responsible for all visitors' safety and health within the site.

Part-III: Occupational Health and Welfare

68. Medical Facilities

68.1 Every effort should be made to provide a healthy environment for workmen. Personal protection should be regarded as the last line of defence, e.g. respiration and ear protectors.

68.2 Physical Fitness of Workmen

68.2.1The contractor shall ensure that his employees/workmen subject themselves to such medical examination as required under the law or under the contract provision and keep a record of the same.

68.2.2The contractor shall not permit any employee/workmen to enter the work area under the influence of alcohol or any drugs.

68.3 Medical Examination

68.3.1The contractor shall arrange a medical examination of all his employees including his sub-contractor employees employed as drivers, operators of lifting appliances and transport equipment before employing, after illness or injury, if it appears that the illness or injury might have affected his fitness and, thereafter, once in every two years up to the age of 40 and once in a year, thereafter.

i)The Contractor shall maintain the confidential records of medical examination or the physician authorised by the Employer / Engineer.

ii)No building or other construction worker shall be charged for the medical examination. The costs of such examinations are borne by contractor.

iii)The medical examination shall include Full medical and occupational history.

a)Clinical examination with particular reference to

i)General Physique;

ii)Vision: - Total visual performance using standard orthorator like Titmus Vision Tester should be estimated and suitability for placement ascertained in accordance with the prescribed job standards.

ii) Hearing: - Persons with normal must be able to hear a forced whisper at twenty-four feet. Persons using hearing aids must be able to hear a warning shout under noisy working conditions.

iii) Breathing: - Peak flow rate using standard peak flow meter and the average peak flow rate determined out of these readings of the test performed. The results recorded at pre- placement medical examination could be used as a standard for the same individual at the same altitude for reference during subsequent examination.

v)Upper Limbs: - Adequate arm function and grip

vi)Spine: - Adequately flexible for the job concerned.

vii)Lower Limbs: - Adequate leg and foot flexibility.

viii)General: - Mental alertness and stability with good eye, hand and foot coordination.

ix)Any other tests which the examining doctor considers necessary.

NOTE: If the contractor fails to initiate and continue medical examinations as stated previously, the Employer / Engineer has the right to appoint a medical examiner through an agency and deduct the cost and overhead charges against the contractor.

68.4 Occupational Health Centre

68.4.1The contractor shall ensure the provision of a site occupational health centre. This may be mobile or static however, both must be maintained in good order and complete with facilities as per the scale laid down in Schedule X of BOCWR. A construction medical officer appointed in an occupational health centre possesses the qualification as laid down in Schedule XI of BOCWR.

68.5 Ambulance van and room

68.5.1The contractor shall ensure the availability of an ambulance at site. First Aid room shall be constructed as per the local laws.

68.6 First-aid boxes

68.6.1The contractor shall ensure a minimum of one First-aid box for every 100 workers is provided and maintained for providing First-aid. Every First-aid box is distinctly marked “First-aid” and is equipped with the articles specified in Schedule III of BOCWR.

68.7 HIV/ AIDS prevention and control

68.7.1The contractor shall appoint agency for the effective implementation of HIV/AIDS prevention and control.

68.8 Prevention of mosquito breeding

68.8.1Measures shall be taken to prevent breeding at site. The measures to be taken shall include:

i)Empty cans, oil drums, packing and other receptacles, which may retain water shall be deposited at a central collection point and shall be removed from the site regularly.

ii)Still waters shall be treated at least once every week in order to prevent mosquito breeding.

iii)Contractor’s equipment and other items on the site, which may retain water, shall be stored, covered or treated in such a manner that water could not be retained.

iv)Water storage tanks shall be provided.

v)Posters in Hindi, Tamil and English, which draw attention to the dangers of permitting mosquito breeding, shall be displayed prominently on the site.

vi)The contractor at periodic intervals shall arrange to prevent mosquito breeding by fumigation / spraying of insecticides. Most effective insecticides shall include SOLFAC WP 10 or Baytex, The Ideal Larvicide etc.

68.9 Alcohol and drugs

68.9.1The contractor shall ensure at all times that no employee is working under the influence of alcohol / drugs which are punishable under Govt. regulations.

68.9.2Smoking at public worksites by any employee is also prohibited.

69. Welfare Measures for workers

69.1 Latrine and Urinal Accommodation

69.1.1The contractor shall provide one latrine seat for every 20 workers up to 100 workers and thereafter one for every additional 50 workers. In addition, one urinal accommodation shall be provided for every 100 workers. Latrine and urinals shall be provided as per Section 33 of BOCWA and maintained as per Rule 243 of BOCWR and shall also comply with the requirements of public health authorities.

69.1.2When women are employed, separate latrine and urinals accommodation shall be provided on the same scale as mentioned above.

69.1.3Latrine and urinals shall be provided and maintained shall also comply with the requirements of public health authorities.

69.1.4In case of mobile e.g. track laying, the zone of work is constantly moving at elevated level or at underground level. In such cases mobile toilets with proper facility to drain the sullage shall be provided at reasonably accessible distance.

69.1.5Should the contractor fail to provide the required number of urinals and latrines or fail to maintain it as per the requirements of Public Health laws, the Employer / Engineer shall have the right to provide/maintain through renowned external agencies at the cost of the contractor.

69.2 Canteen:

69.2.1The contractor shall provide an adequate canteen and the charges for food stuffs shall be based on 'no profit no loss' basis. The price list of all items shall be conspicuously displayed in such canteen.

69.3 Serving of tea and snacks at the workplace:

69.3.1The building or other construction work where a workplace is situated at a distance of more than 200 m from the canteen, the contractor shall make suitable arrangement for serving tea and light refreshments to their employees.

69.4 Drinking water

69.4.1The contractor shall make in every worksite, effective arrangements to provide sufficient supply of wholesome drinking water with minimum quantity of 5 litres per workman per day. Quality of the drinking water shall conform to the requirements of national standards on Public Health.

69.4.2Due care shall be taken to ensure drinking water facilities are easily accessible to the workforce and within a distance of 200m from the place of work.

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69.4.3 All such points shall be legible marked “Drinking Water” in a language understood by a majority of the workforce. Drinking water points are not to be located within six metres of any washing places, urinals or latrines.

69.5 Labour Accommodation

69.5.1 The contractor shall provide free of charge and as near as possible to the workplace, temporary living accommodation for all workers. All provided accommodation shall conform to the provisions made under Section 34 of BOCWA. These accommodations shall have cooking place, bathing, washing and lavatory facilities.

70. Permissible exposure limits

70.1.1The Exposure to air Borne contaminants of an employee working in your tunnel or shaft should not exceed the exposure limit.

70.1.2Employees should be removed from any area in which there is an airborne contaminant at a concentration which exceeds the exposure limit for that contaminant.

70.1.3Portable instruments should be provided to test the atmosphere quantitatively for carbon monoxide hydrogen sulphide Nitrogen dioxide flammable or toxic gases dust or other toxic contaminants that occur in the Tunnel or shaft. Tests should be conducted before each shift and once in every shift or more frequently you are necessary to ensure that the required quality of air is maintained a record of all tests should be maintained and be kept available for inspection.

71. Dust

71.1.1The amount of dust generated should be minimized. All dusts are harmful and exposure should be kept within national limits.

71.1.2The main consequences from dust exposure are pneumoconiosis and silicosis. The latter can occur after long exposure, where there is considerable quartz content in the airborne dust.

71.1.3In its mildest form dust exposure annoys by making sensitive parts of the respiratory tract very uncomfortable. In addition, dust reduces visibility and enters the respiratory tract.

71.1.4Exposure to some dusts e.g. cement, can cause dermatitis.

72. Protection Against Dust

72.1.1West flushing (preferred) or dust extraction should be used during drilling.

72.1.2The working face, tunnel walls and muck pile should be watered for at least 15 minutes after every blast and before mucking operations begin. The muck pile should also be watered during mucking operations.

72.1.3Dust extraction units or water sprays should be placed at sources of dust emission such as on tunnel boring machines, conveyor transfer points, shotcrete mixing units et.,

72.1.4If the above measures are unsuccessful at controlling airborne dust concentrations, additional ventilation should be provided.

72.1.5Dust masks for individual protection should only be necessary to use in exceptional cases and for short duration work at that.

73. Gaseous Contamination

73.1.1Toxic gases are generated through the use of explosives, diesel engines, from ground contamination or, more rarely, from natural sources.

73.1.2Sufficient ventilation should be provided prevent the buildup of a toxic or potentially explosive atmosphere in the tunnel.

71.1.3Gases from blasting operations are basically oxides of nitrogen, ammonia and oxides of carbon. Therefore, it is necessary to provide powerful ventilation in order to reduce the re-entry time to a minimum. The characteristics of the fume emitted should be considered in the selection of the explosives.

74. Emissions from Power Units

74.1.1Petrol driven internal combustion engines shall not be used underground because of their large emissions of CO (10% of exhaust gas) and the fire and explosion hazard.

74.1.2Only low emission or “clean burn” diesel engines or electrically driven or battery operated motors should be used.

74.1.3Engines should be properly adjusted and maintained to operate efficiently under prevailing conditions in order to reduce toxic exhaust emissions to the absolute minimum.

74.1.4Engines should be effectively silenced so as to achieve a noise level not exceeding LAeq 85 Db(A) as measured in the open, at a distance of 1 m.

74.1.5Exhaust systems should be fitted with an efficient catalytic converter and particulate filter.

74.1.6Direct injection air cooled engines are preferred for reason of better exhaust emission characteristics.

74.1.7Efficient air intake cleaners should be provided.

74.1.8 Idling of diesel engines should be kept to a minimum.

74.1.9 Storage of diesel fuel underground should be prohibited.

74.1.10All vehicles underground, should have efficient fixed extinguishing systems.

74.1.11Atmospheric monitoring in the vicinity of any engine should be carried out at least once per month, with the engine running a full load and when idling.

74.1.12Exhaust gas analyses should be taken at least once every three months under full load and idling conditions.

75. Heat

75.1.1Cooling the tunnel atmosphere should be undertaken when the temperature routinely exceeds 27*C.

75.1.2Cooling may consist of increasing the ventilation flow and / or installing cooling equipment.

Appendix 1

An Example of the Identification Badge

Colour	:	YELLOW
Size	:	35 mm x 40 mm
Material	:	Plastics (or hard board) with lamination and 5 mm hole for string.

FRONT

Compressed Air Worker

Name:

In case of sudden illness or emergency of unknown or doubtful cause, the wearer of this badge should be rushed to

MEDICAL LOCK

XXXXXXXXXXXXXXXXXXXXXXX

(……………Site Address…………………)

Tel: XX-XXXXXXXXXX

REVERSE

Compressed Air Worker

Name:

In case of sudden illness or emergency of unknown or doubtful cause, the wearer of this badge should be rushed to

MEDICAL LOCK

XXXXXXXXXXXXXXXXXXXXXXX

(……………Site Address…………………)

Tel: XX-XXXXXXXXXX

BUREAU VERITAS (INDIA) PRIVATE LIMITED MUMBAI